Single-Cell Dissection of Healthy vs Malignant Bone Marrow Microenvironments
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***Abstract*—This study conducts a comparative single-cell RNA sequencing (scRNA-seq) analysis of healthy and malignant bone marrow microenvironments to investigate cellular heterogeneity and intercellular communication. Leveraging the Seurat package in R, we preprocess, normalize, and annotate two distinct datasets. Through quality control and initial clustering, we lay the foundation for downstream analysis with CellChat. Our preliminary results reveal clear distinctions in mitochondrial content, gene expression complexity, and cell type distribution between healthy and malignant samples, highlighting potential avenues for understanding tumor-associated signaling.**

I. INTRODUCTION

The bone marrow is a dynamic and complex tissue responsible for producing blood cells through hematopoiesis. It consists of various cell types, including hematopoietic stem and progenitor cells, immune cells, and stromal components that form a tightly regulated network. In diseases like leukemia and multiple myeloma, malignant cells hijack the bone marrow niche, altering its architecture and communication networks to create a supportive environment for cancer progression.

Single-cell RNA sequencing (scRNA-seq) has emerged as a powerful technique to dissect tissue heterogeneity and cellular states at unprecedented resolution. It enables the identification of cell subpopulations, and their gene expression profiles, providing a window into the underlying cellular architecture of both healthy and diseased tissues. In parallel, computational tools such as **Seurat** and **CellChat** allow for systematic analysis of cell clustering and intercellular communication.

This project aims to perform a comparative scRNA-seq analysis of healthy and malignant bone marrow environments. We use Seurat for data preprocessing, and dimensionality reduction, followed by SingleR for cell type annotation, and CellChat to infer and compare cell-cell signaling pathways. This approach offers a comprehensive view of the cellular landscape and communication dynamics altered in malignancy.

II. METHODS

1) Download and prepare the data

a) Data Acquisition and Setup

Data were obtained from the GEO GSE176131 database (GSM5356722 for healthy, GSM5356726 for malignant). Expression matrices, barcodes, and feature annotations were downloaded in the 10X Genomics format and organized into two directories: *mtx_data/control* and *mtx_data/malignant*.

b) Software and Packages

The analysis was conducted in R using a suite of packages including **Seurat**, **SCpubr**, **scater**, **clusterProfiler**, **SingleR,** and **CellChat**.

c) Data Import and Merging

Data were read using Read10X() and transformed into Seurat objects. To retain sample identity, cell barcodes were renamed "Helathy" and "Malignant". This allowed tracking of origin throughout the analysis.

2) Quality Control

Three key metrics were used for cell-level QC:

- **Library Size** (nCount_RNA)
- **Feature Count** (nFeature_RNA)
- **Mitochondrial Gene Content (percent.mt)**

Violin plots and histograms, stratified by condition, enabled visual inspection of each metric's distribution. Initially, outliers were identified using the median absolute deviation (MAD)-based method. However, this approach excluded a substantial number of cells. Therefore, thresholds were manually adjusted based on the distributions observed in the histograms and violin plots to better reflect the characteristics of the dataset.

Cells were retained for downstream analysis if they satisfied the following criteria:

- RNA counts between 300 and 50,000
- Detected gene features between 200 and 6,000
- Mitochondrial gene content below 15%

These thresholds ensured a balance between data quality and cell retention.

Gene-Level Quality Control:

Following cell filtration, a second QC step was performed at the gene level. Genes that were expressed (i.e., had non-zero counts) in fewer than 3 cells were excluded. This step eliminates genes with extremely sparse expression patterns, which are generally uninformative and can introduce noise in downstream analyses.

A custom filtering function was used to identify and retain only those genes expressed in at least 3 cells. This reduced the number of genes from 33,538 to 18,546, retaining genes with reliable expression across multiple cells.

3) Normalization, Variable Feature Selection, and Data Scaling

To prepare the data for downstream analysis, we first performed three essential preprocessing steps using the Seurat package:

a) Log-Normalization

We applied Seurat's NormalizeData() function, which performs a log-normalization of gene expression counts for each cell. This method normalizes gene expression measurements by the total expression, multiplies it by a scale factor (default: 10,000), and then applies a log-transformation. This helps to stabilize the variance across genes and cells.

b) Variable Feature Selection

Using the FindVariableFeatures() function, we identified the most variable genes across cells. These highly variable genes capture biological heterogeneity and are prioritized in downstream dimensionality reduction and clustering. Seurat selects these based on dispersion (variance/mean ratio) and mean expression.

c) Data Scaling

The ScaleData() function was applied to center and scale the data. Each gene's expression was z-score normalized across all cells (i.e., the expression values were mean-centered and divided by the standard deviation). This step ensures that genes contribute equally during PCA and clustering.

4) SCTransform Normalization

To further enhance data quality and address technical variation, we employed SCTransform, a variance-stabilizing transformation that models gene expression using a regularized negative binomial regression. The transformation accounts for sequencing depth and other confounding variables.

Using SCTransform(), we:

- Normalized the data,
- Identified the top 3,000 variable features,
- Scaled the data, and
- Replaced the default "RNA" assay with a new SCTransform-based "SCT" assay.

5) Visualization of Variable Features

To visualize gene variability after SCTransform, we identified the top 10 most variable genes using VariableFeatures() . These were plotted using VariableFeaturePlot() and labeled with LabelPoints() for emphasis. The plot was exported as a high-resolution figure titled "Variable Features Top10.png" for documentation and further analysis.

6) Cell Cycle Scoring

We used Seurat's 2019-curated lists of canonical.

The Seurat function CellCycleScoring() was applied with the marker lists specified via s.features and g2m.features. For every cell the function computes:

- **S.Score**: mean normalized expression of S-phase genes.
- **G2M.Score**: mean normalized expression of G2/M-phase genes.

A cell is labelled **S** if S.Score > G2M.Score and at least one score exceeds background. A cell is labelled G2M if G2M.Score > S.Score under the same condition.

Cells with low values for both scores default to G1 (or quiescent G0). Setting set.ident = TRUE records the categorical label (Phase) as the Seurat identity class and appends both continuous scores to the object metadata.

7) Dimensionality Reduction

a) Principal Component Analysis (PCA)

PCA is a linear dimensionality reduction technique that projects the expression matrix into a set of orthogonal components (principal components or PCs), It allows for the detection of major patterns in the data, such as batch effects, technical variation, or biological signals including cell cycle phase or condition (e.g., healthy vs malignant).

PCA was applied to the normalized and scaled gene expression matrix stored in the Seurat object using RunPCA(). The dimensionality-reduced embeddings (PC_1 vs PC_2) were visualized using a custom do_DimPlot() function.

Phase-based PCA Plot: Cells were colored based on their cell cycle phase (G1, S, G2/M), as previously annotated using CellCycleScoring(). This plot helps visualize whether cell cycle variation is a major contributor to variance in the dataset.

Condition-based PCA Plot: Cells were also visualized according to the experimental condition (healthy vs malignant) to examine biological separation in PC space.

Two PCA plots were generated and saved: "Cell Cycle Phase PCA.png" visualizes the distribution of cells across principal components by **cell cycle phase**.

"Condition PCA.png" visualizes the separation of cells based on the **experimental condition**.

b) Assessing Technical Variation in PCA Space

To ensure that technical factors such as library size, feature counts, and mitochondrial gene content do not dominate the principal components (PCs), we projected these metrics onto the PCA space using do_FeaturePlot().

Technical Covariates Assessed:

Library Size (nCount_RNA): Total RNA molecules detected per cell.

Number of Features (nFeature_RNA): Number of unique genes detected per cell.

Mitochondrial Gene Percentage (percent.mt): Proxy for cell health and stress.

Each metric was visualized across the first two principal components (PC1 and PC2), split by the experimental condition (healthy vs malignant). The use of a reversed Viridis palette enhanced visibility of gradients across the PCA embedding.

c) Assessing Biological Condition in PCA Space

To determine if the experimental condition (healthy vs malignant) contributed significantly to variation in gene expression, PCA embeddings were plotted across several PC pairs (PC1–PC6), grouping cells by their condition.
This step was crucial for:

- Understanding how early PCs relate to biological conditions.
- Selecting PCs relevant for downstream clustering or visualization.
- Avoiding confounding effects during interpretation of major sources of variability.

d) Regressing Cell Cycle Effects

Since cell cycle phase was a potential source of biological variation not of interest in this analysis, we removed its influence by regressing out the Phase metadata variable using the SCTransform() function.
After regression:

- PCA was re-run using variable features identified by SCTransform.
- Updated PCA plots were generated to visualize condition-based structure post-regression, confirming successful mitigation of cell cycle-related variation.

e) Elbow Plot for PC Selection

To determine the optimal number of principal components to retain for downstream analysis (e.g., UMAP, clustering), we generated an Elbow Plot using Seurat's ElbowPlot() function. The plot shows the standard deviation (variance explained) for each PC and helps identify the point where the marginal gain in explained variance drops off.
The elbow point was manually determined, and the first **20 PCs** (pcs_use <- 1:20) were selected for subsequent steps in dimensionality reduction and clustering.

7) UMAP Dimensionality Reduction

To visualize high-dimensional data in two dimensions, Uniform Manifold Approximation and Projection (UMAP) was applied to the PCA-reduced matrix using the top 20 principal components (pcs_use). UMAP was selected for its ability to preserve local neighborhood structure while capturing global variation, which aids in both clustering and biological interpretation.
UMAP plots were generated to visualize:

- Condition: to inspect separation between healthy and malignant cells.
- Cell Cycle Phase: To verify that cell cycle is no longer a driver of variation.

8) Differential Expression Analysis

To identify differentially expressed genes (DEGs) between Malignant and Healthy cell populations, we performed a Wilcoxon rank-sum test using the FindMarkers() function from the Seurat package. The analysis was conducted on the subset Seurat object seu_subset, grouped by the "condition" metadata column.
The test parameters were set as follows:

- ident.1 = "Malignant" and ident.2 = "Healthy" to contrast malignant cells against healthy ones.
- test.use = "wilcox" for non-parametric statistical testing.
- logfc.threshold = 0.25 to include genes with a minimum log2 fold change of ±0.25.
- min.pct = 0.10 to retain genes expressed in at least 10% of cells in either group.

The resulting table of DEGs was saved to write/tables/Markers.csv. Each gene was annotated with a "regulation" column indicating whether it was upregulated ("Up") or downregulated ("Down") in the malignant group based on the sign of avg_log2FC.
To visualize the results, two types of plots were generated:

1. Volcano Plot: Created using EnhancedVolcano() to highlight genes with both statistical significance (p_val_adj < 0.05) and substantial expression changes (|avg_log2FC| > 0.25). This plot was saved as write/figures/Volcano Plot.png.
2. Heatmap: A heatmap was generated using DoHeatmap() to display the top 20 most upregulated genes, selected by sorting DEGs in descending order of avg_log2FC. The heatmap grouped cells by "condition" and was saved as write/figures/DEA Heatmap.png.

9) Clustering Analysis

To discover transcriptionally distinct subpopulations, we applied graph-based clustering using the Louvain algorithm implemented in Seurat.
The dataset was split by condition into healthy_obj and malignant_obj subsets. For both subsets the nearest-neighbor graphs were constructed using PCA, Clustering was performed at resolutions from 0.0 to 1.0 in 0.1 increments to explore hierarchical cluster structures.
Clustering Tree Analysis:
To determine the optimal clustering resolution clustering trees were generated using the Clustree package, these trees visualize how clusters split/merge across resolutions and incorporate SC3 stability scores to assess cluster robustness.
Separate plots were generated for:

- Healthy cells (saved as healthyClustrees.png)
- Malignant cells (saved as Malignant_Clustrees.png)

Based on tree topology and stability, resolution 0.4 was chosen for both conditions.
UMAP projections were generated for both healthy and malignant datasets, colored by seurat_clusters, to visually assess the spatial distribution and identity of clusters in two dimensions.

10) Feature and Expression Visualization

a) Feature plot

The top variable genes identified in the subset were plotted using UMAP-based feature plots (FeaturePlot) to observe expression trends spatially across the dataset.

b) Violin plot

Violin plots (VlnPlot) were generated to visualize expression levels of variable genes across conditions, highlighting key biological differences.

11) Cell Type Annotation (SingleR)

Annotation was performed separately for both healthy and malignant Seurat objects using the Human Primary Cell Atlas reference from celldex.

Healthy Cells:

a) Cluster-based annotation:

For each dataset, SingleR assigned cell types at the cluster level using average expression per cluster.

Marker genes were computed for each cluster using FindAllMarkers() with a logFC threshold of 0.05 and minimum expression in 15% of cells.

The pruned.labels output from SingleR was used to filter out ambiguous assignments.

Annotated cell types were added back to the Seurat object under the metadata column annotated_clusters.

b) Cell-Based annotation:

Individual cells were also annotated directly to increase granularity. Low-confidence annotations (NA) were filtered out.

Annotated labels were stored as metadata (annotated_clusters and annotated_cells) in the Seurat objects and saved as RDS files for future use.

Malignant Cells:

a) **Cluster-based annotation:**

Similarly to the healthy dataset:

- Marker genes were identified for each cluster (logFC > 0.05, min.pct > 15%).
- Average expression profiles were passed to SingleR to infer cluster-level cell types.
- Pruned labels were used to populate the annotated_clusters metadata field.

b) **Cell-Based annotation:**

The entire malignant expression matrix was passed to SingleR in per-cell mode, resulting pruned annotations were stored in the annotated_cells field. Cells with low-confidence or ambiguous matches were excluded.

c) **Filtering for Shared Cell Types**

To ensure valid downstream comparisons, only **shared cell types** between healthy and malignant datasets were retained.

This filtering step eliminated unique or unassignable populations from further analyses to control for confounding differences due to annotation mismatches.

12) CellChat Analysis

a) Object Creation and Preprocessing

Two CellChat objects (chat_healthy, chat_malignant) were created from the healthy and malignant Seurat objects using createCellChat() with group.by = "annotated_cells". The human ligand-receptor database CellChatDB.human was assigned to both objects. The expression data was subset to include only genes involved in secreted signaling.

b) Communication Probability Inference

For both healthy and malignant samples:

- identifyOverExpressedGenes() and identifyOverExpressedInteractions() were used to detect highly expressed ligand-receptor interactions.
- computeCommunProb(type = "triMean") calculated communication probabilities.
- filterCommunication(min.cells = 10) removed low-cell-count groups.
- computeCommunProbPathway() inferred pathway-level communication.
- aggregateNet() summarized the communication network.

Excluded low-cell groups:

- Healthy: MEP, Pro-B_cell_CD34+
- Malignant: BM, Platelets, Pro-B_cell_CD34+

Each object was saved as an RDS checkpoint.

c) Network Centrality Analysis

We computed network centrality measures using netAnalysis_computeCentrality() on the netP slot for both samples, identifying key sender, receiver, mediator, and influencer cell types in the network.

d) Merging and Visualization

The healthy and malignant CellChat objects were merged using mergeCellChat(), enabling comparative visualizations across conditions. The merged object and the object list were saved for reuse.

- **Total Interaction Counts & Weights:**
  compareInteractions() was used to visualize total counts and weights of interactions in both conditions.The plot was saved as a PNG image (total_interaction_counts_weights.png).
- **Differential Interaction Networks:**
  netVisual_diffInteraction() was used to compare interaction networks by count and by weight across conditions.
- **Heatmaps:**
  Heatmaps of interaction count and weight were created using netVisual_heatmap() and combined using patchwork and ComplexHeatmap.

- **Circle Plots – Interaction Count and Strength:**
  netVisual_circle() visualized network topology based on interaction number and strength for each condition separately. The maximum weights were scaled uniformly using getMaxWeight().
- **Signaling Role Scatter Plot:**
  netAnalysis_signalingRole_scatter() visualized the functional roles (sender/receiver) of each cell type. Plots were aggregated with wrap_plots() and saved.
- **Bubble Plots for Selected Pathways:**

For selected immunomodulatory pathways (MIF, GALECTIN, ANNEXIN), netVisual_bubble() visualized ligand-receptor pairs in both conditions. Each plot was saved with pathway-specific filenames.

- **Hierarchy Plots:**
  For the same three pathways, netVisual_aggregate() was used to generate hierarchy plots. Weights were scaled across datasets to ensure comparability.
- **Gene Expression Violin Plots:**
- plotGeneExpression() plotted pathway gene expression split by dataset (Healthy vs Malignant). A subtitle clarified color mapping.
  Each plot was saved under a pathway-specific filename.

13) Data Management

All intermediate and final R objects were saved using saveRDS() and reloaded as needed using readRDS(). All plots were saved in high-resolution PNG format to write/figures/, and CSV tables were written to write/tables/.

III. DISCUSSION & RESULTS

1) QC results

Preliminary QC results highlight biologically relevant differences between healthy and malignant samples:

a) **Library Size**:

Violin plot Interpretation: (Figure1)

A violin plot comparing the library sizes across conditions (Figure 1) revealed that the majority of cells in both the healthy and malignant samples had UMI counts concentrated below 25,000. However, a distinct difference in distribution was observed. The malignant sample exhibited a slightly elevated median library size compared to the healthy sample, along with a longer upper tail, indicating a higher proportion of cells with exceptionally large transcript counts. These high-UMI cells may correspond to transcriptionally hyperactive populations such as malignant plasma cells or activated immune cells.

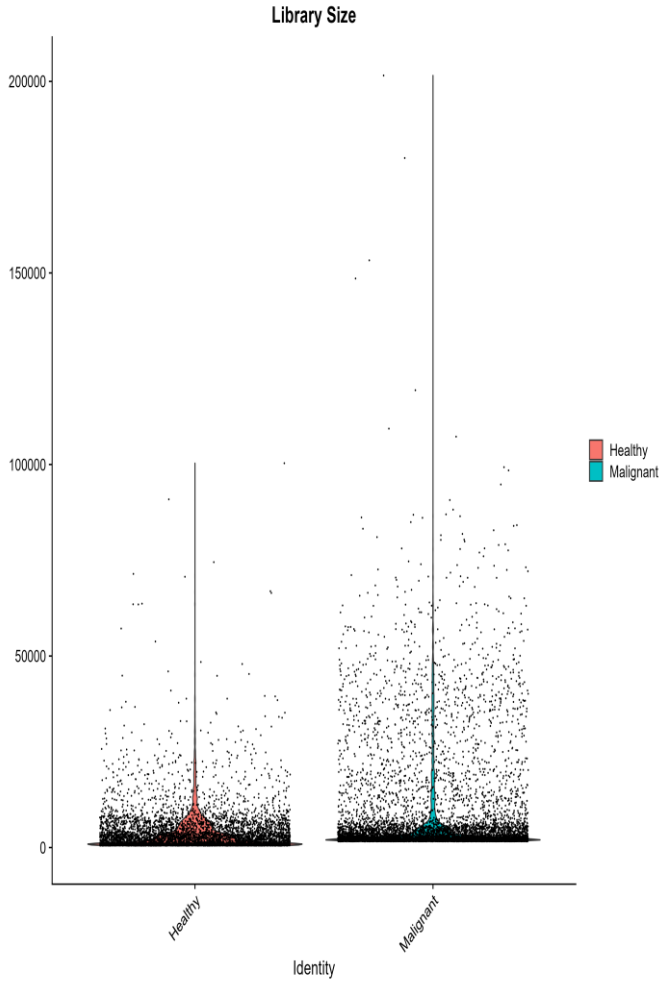


Figure1; *Library Size Violin Plot*

Histogram Analysis: (Figure2)

Further characterization of library size distributions using histograms (Figure 2) demonstrated that the healthy sample had a tightly clustered distribution, with most cells falling between 1,000 and 10,000 UMIs and very few exceeding 20,000. In contrast, the malignant sample showed a broader, right-skewed distribution, with a long tail extending beyond 200,000 UMIs. This suggests increased transcriptional heterogeneity in the malignant condition.

Vertical threshold lines were used to indicate quality control (QC) boundaries, with a lower limit of 300 UMIs to exclude low-quality droplets, and an upper limit of 50,000 UMIs to filter potential multiplets or outliers. These thresholds successfully encompassed the majority of cells while excluding atypical extremes.

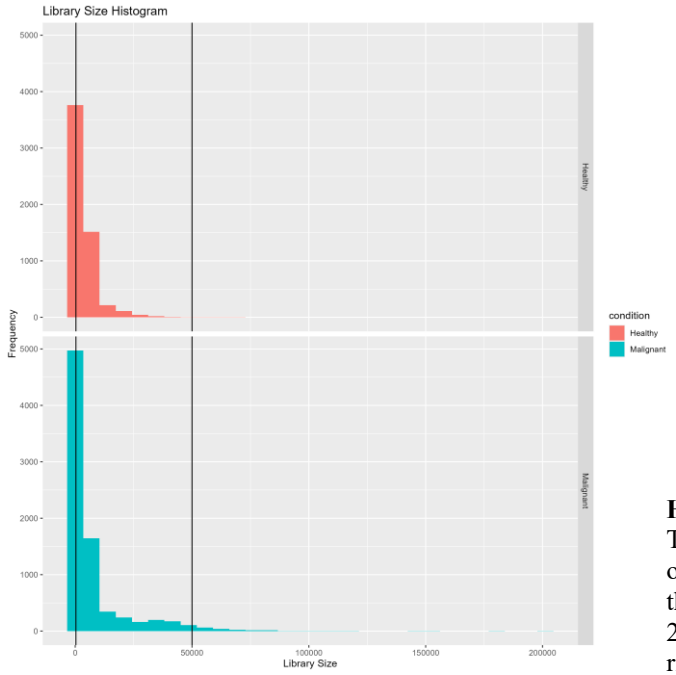


Figure2; *Library Size Histogram*

b) Gene Feature Distribution Across Conditions:

To evaluate transcriptomic complexity across conditions, we analyzed the number of detected genes (total features) per cell using both violin plots and histograms for the healthy and malignant bone marrow samples (Figures 3,4).

Violin Plot Overview (Figure3)

The violin plot(figure3) illustrates the distribution and density of total features per cell. In the healthy sample, most cells displayed a relatively compact distribution, with the majority expressing between 500 and 2,500 genes. The distribution showed a modest tail extending toward higher values, but few cells exceeded 5,000 features.

In contrast, the malignant sample exhibited a broader and more right-skewed distribution. While the majority of cells still clustered within the lower-to-mid range (500–2,500 genes), a substantial number of cells exceeded 5,000, with some reaching up to 9,000–10,000 genes. This indicates a subset of cells with significantly increased gene expression complexity, possibly corresponding to transcriptionally active malignant clones or plasma cells.

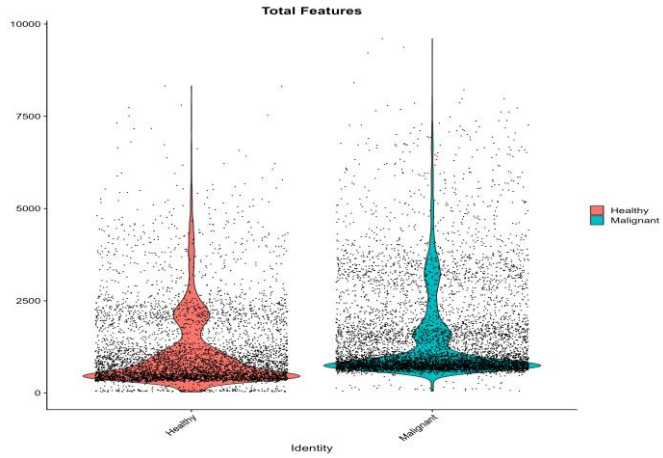


Figure3; *Total Features Violin Plot*

Histogram Insights and QC Thresholds (Figure4)

The accompanying histogram (figure4) reinforced these observations by visualizing feature count frequencies across the dataset. The healthy sample displayed a steep drop-off after 2,500 features, **whereas the malignant sample** retained a long right tail, with more high-feature cells distributed beyond 5,000.

Vertical lines in the histogram mark the quality control thresholds, which are set at:

- Lower threshold (200 features): to filter out low-quality or dying cells with minimal transcriptional output.
- Upper threshold (6,000 features): to exclude likely doublets or multiplets with abnormally high feature counts.

These thresholds effectively capture the majority of high-quality single-cell transcriptomes while excluding technical artifacts and biologically implausible outliers.

Biological Implications

The broader distribution and elevated feature counts in the malignant condition reflect greater transcriptional heterogeneity and complexity. This is consistent with the presence of diverse malignant subclones, active immune components, or other supporting stromal populations within the tumor microenvironment. In contrast, the healthy bone marrow shows a more uniform gene expression profile, suggesting a more stable and regulated transcriptional landscape.

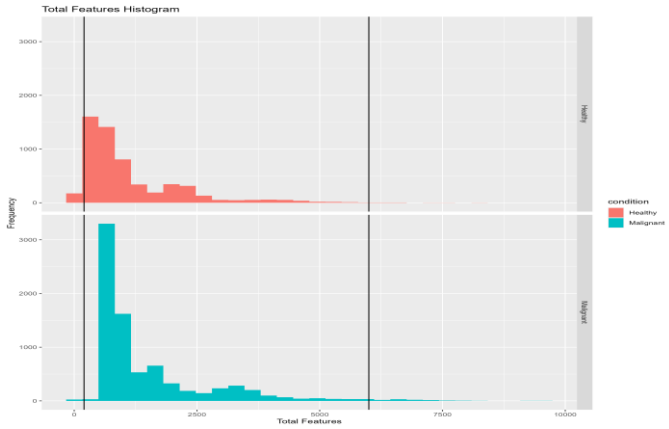


Figure4; *Total Features Histogram*

c) Mitochondrial Content:

The percentage of mitochondrial gene expression per cell was analyzed as a key quality control metric to detect stressed, apoptotic, or dying cells. We visualized this metric across healthy and malignant bone marrow samples using both histogram and violin plots (Figures 5,6).

Violin Plot Interpretation: (Figure5)

The violin plot (Figure 5) illustrates the distribution of mitochondrial gene percentages within each sample. The majority of cells in both conditions exhibited low mitochondrial content, typically below 10–15%. However, the malignant sample displayed a broader distribution, with more cells having intermediate-to-high mitochondrial content compared to the healthy sample.

Outliers with extremely high mitochondrial percentages (up to 100%) were present in both samples, although these were more frequently observed in the malignant dataset. These cells likely represent damaged or dying cells and are typically excluded during QC filtering.

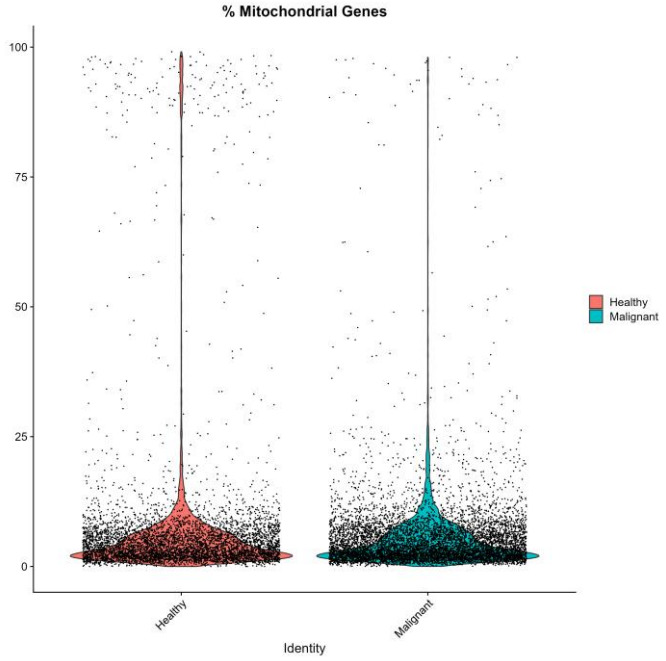


Figure5; *Percentage Mitochondrial Genes Violin Plot*

Histogram Analysis and Thresholding: (Figure6)

The histogram (Figure6) provided a clearer view of mitochondrial content frequencies:

- In the healthy sample, most cells had <15% mitochondrial gene expression, forming a sharp left-skewed distribution. A small fraction of cells showed abnormally high mitochondrial content, possibly due to sequencing artifacts or damaged cells.
- In the malignant sample, although the majority of cells still fell under 10-15%, the distribution was more dispersed, with a longer tail extending toward higher percentages.

A vertical black line on both histograms marked the mitochondrial QC threshold (15%), commonly used to filter out cells under oxidative or apoptotic stress. This threshold effectively excludes most outlier cells while preserving high-quality transcriptomes.

Biological Implications

While both samples contain some high-mitochondrial-content cells, the broader distribution in the malignant group may reflect:

- Increased cellular stress within the tumor microenvironment,
- Metabolic alterations associated with malignant transformation,
- Or the presence of dying tumor or immune cells.

The consistency of mitochondrial gene content across most cells suggests that the dataset is overall of good quality, with only a subset requiring exclusion.

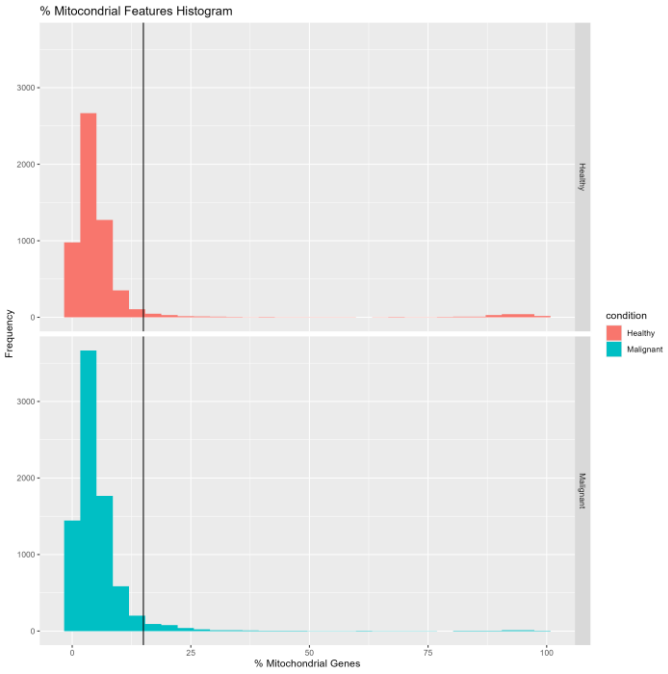


Figure6; *Percentage Mitochondrial Genes Histogram*

d) Integrated Quality Control Metrics

To jointly assess cell quality across samples, we visualized total UMI counts (library size), number of detected genes (features), and mitochondrial gene expression percentage in a single scatter plot (Figure 9). Each point represents a cell, colored by its percentage of mitochondrial gene content, with black lines marking quality control thresholds.

Interpretation of Axes and Thresholds (Figure7)

- The x-axis represents total UMI counts per cell (library size).
- The y-axis shows the number of genes detected per cell (total features).
- Color intensity indicates the percentage of mitochondrial gene content, ranging from dark purple (0%) to bright yellow (100%).
- Vertical threshold (50,000 UMIs) and horizontal threshold (6,000 features) are used to identify potential doublets/multiplets.

- A low-end threshold for both axes (300 UMIs and 200 features) excludes low-quality or dying cells.

Healthy Sample (Top Panel)

Most cells cluster tightly below the upper QC thresholds, with a clean arc-like trajectory suggesting a strong positive relationship between UMI count and feature count. Mitochondrial content is generally low (dark purple), with a small number of outliers exhibiting high mitochondrial expression, visible as yellow dots. A few cells exceed the feature or UMI thresholds and are likely multiplets or highly active plasma cells.

Malignant Sample (Bottom Panel)

A broader distribution is observed in the malignant sample. While many cells follow the same arc-like pattern as the healthy sample, a notable subpopulation appears with:

- Elevated feature counts and/or library sizes, possibly indicating malignant clonal expansion or high RNA content.
- Slightly higher mitochondrial content on average, with several cells showing elevated levels (>25%), suggesting stress or apoptosis.

This broader spread and greater variation in mitochondrial percentages reinforce the increased heterogeneity and metabolic complexity in the malignant bone marrow microenvironment.

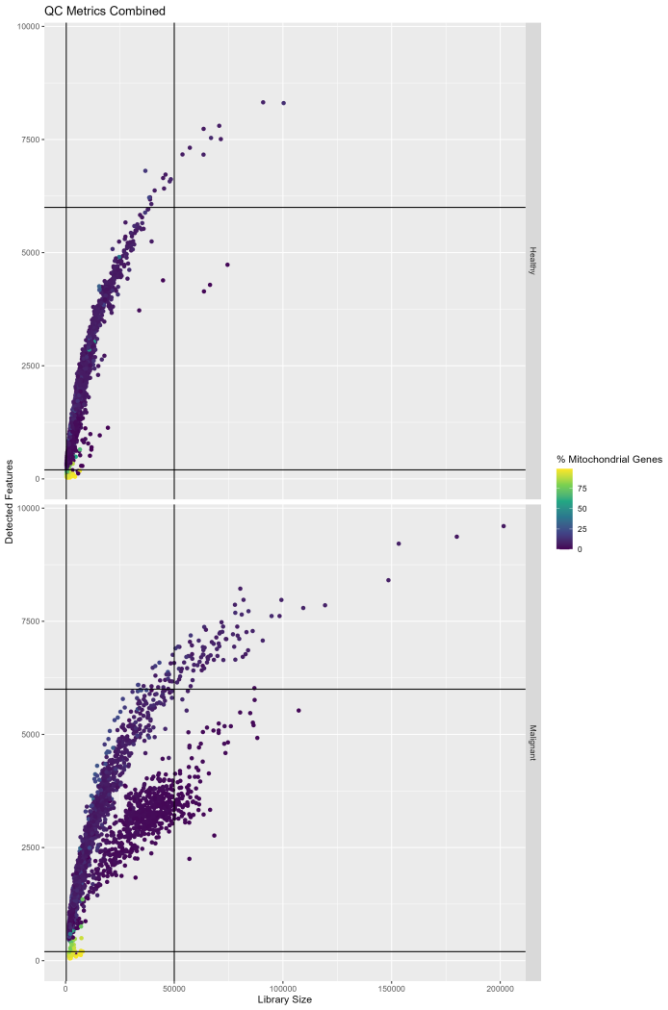


Figure7;*QC Metrics Scatter Plot*

2) Normalization HVG Detection, and Cell Cycle Scoring:

To assess and mitigate the impact of cell cycle–related transcriptional variability, we performed cell cycle scoring using the CellCycleScoring() function from the Seurat package. This step is crucial, as cell cycle heterogeneity can obscure true biological differences, especially when comparing healthy and malignant samples.

Before scoring, the dataset was normalized using NormalizeData(), followed by selection of highly variable features via FindVariableFeatures(), and then scaled with ScaleData(). To further stabilize variance and remove technical effects, we subsequently applied SCTransform(), which replaces the default RNA assay with an SCT assay and yields 3,000 highly variable genes. The SCTransform-derived residuals were used for all PCA, UMAP, and clustering analyses described below. The CellCycleScoring() function then assigned each cell to one of the three canonical phases of the cell cycle: G1, S, or G2M, based on their expression of the phase-specific gene sets.

The resulting distribution of cells across cell cycle phases was as follows:

- G**1**: 7,929 cells
- S: 2,528 cells
- G**2/M**: 2,304 cells

This distribution reflects a biologically plausible scenario for an unsynchronized single-cell population, where the majority of cells are quiescent (G1), and a substantial fraction are actively cycling (S and G2M phases). The presence of a considerable number of proliferative cells (~45%) may reflect the high cellular turnover typical of bone marrow, and potentially higher proliferation in malignant samples.
Identifying and annotating cell cycle phases enables informed regression of this variation in downstream analyses, such as clustering or dimensionality reduction. Without this step, clusters may inadvertently reflect differences in cell cycle stage rather than true biological identities, such as distinct stromal or immune subtypes.

3) Visualizing Variable Features

Identification of Highly Variable Genes (Figure8)

To identify genes exhibiting substantial biological variability across the dataset, we applied variance-based feature selection to detect highly variable genes (HVGs). The results were visualized in a mean–variance scatter plot (Figure 8), where each gene is plotted based on its geometric mean expression (x-axis) and residual variance (y-axis) after normalization.

Overview of Feature Variability

A total of 3,000 variable genes were identified (red points), representing those with the highest residual variance after accounting for expression level.
The remaining 15,023 genes (black points) were considered non-variable and exhibit low variance across cells.
Variable genes are typically retained for downstream steps such as PCA, clustering, and differential expression analysis, as they are most likely to capture biologically relevant heterogeneity.

Top Variable Genes

The plot (Figure8) highlights several of the most variable genes, many of which are known to be associated with immune function or **lineage-specific expression:**
DEFA3 (Defensin Alpha 3) and IGLC2 exhibited the highest residual variance, suggesting strong cell-type-specific expression, likely linked to neutrophils and plasma cells, respectively.

Other highly variable genes include:

JCHAIN – commonly expressed in plasma cells,
CAMP, MPO, PRTN3, ELANE, GNLY, OLFM4, CXCL8 – involved in neutrophil activation, granule formation, and inflammation.
The enrichment of immune-related genes among the top HVGs supports the presence of distinct immune cell subsets in the dataset, as expected in bone marrow tissue.

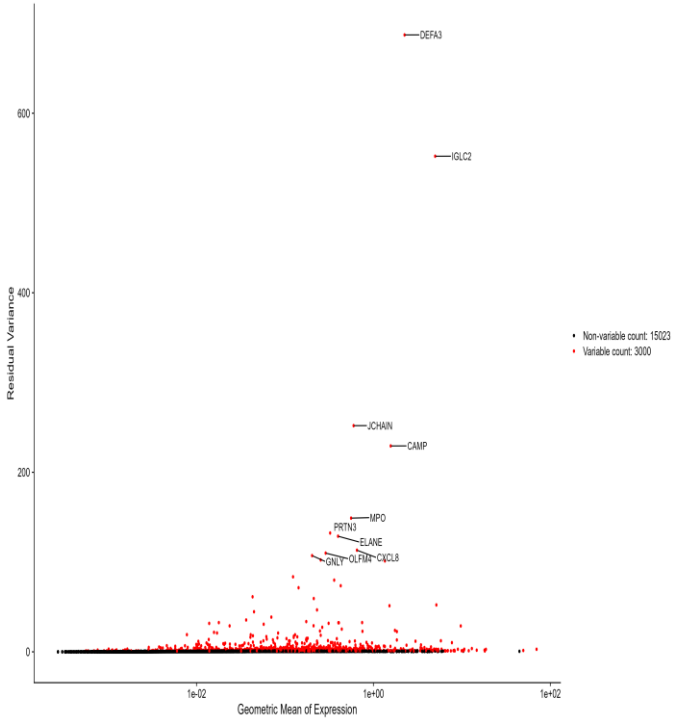


Figure8; Variable Features Top10

4) Dimensionality Reduction and Clustering:

a) Principal Component Analysis (PCA)

To explore transcriptional variation in the dataset, Principal Component Analysis (PCA) was performed following normalization and variable gene selection. The resulting PCA embeddings are visualized in (Figures9,10), with cells colored by either cell cycle phase or experimental condition.

Cell Cycle Phase Projection (Figure 9)

In the first PCA plot, cells are colored by their assigned cell cycle phase (G1, S, or G2M), as inferred from canonical cell cycle marker gene expression.

- Cells in the **G1 phase** (red) form the majority of the dataset and occupy a broad region of the PCA space.
- Cells in the **S phase** (blue) and **G2M phase** (green) are interspersed, with modest enrichment in certain regions, indicating localized transcriptional programs associated with DNA synthesis and mitosis.
- The separation of cycling vs. non-cycling cells is not sharply distinct, suggesting that cell cycle effects are present but not dominating the principal components of transcriptomic variation.

This distribution supports the need for cell cycle regression in downstream steps such as clustering and differential expression, particularly if cell proliferation varies across conditions.

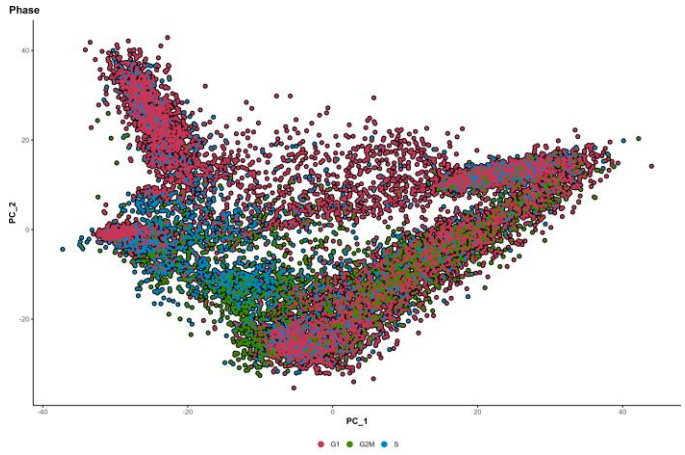


Figure 9; Cell Cycle Phase PCA

Condition-Based Projection (Figure 10)
In the second PCA plot, cells are colored by sample origin **(Healthy in red, Malignant** in green).
Cells from both conditions partially overlap but also show distinct clustering in some regions, indicating condition-specific transcriptional differences.
The malignant cells occupy a slightly broader range in PCA space, consistent with the increased transcriptional heterogeneity observed in previous QC and variance analyses.
The partial overlap suggests the presence of shared cell populations (e.g., conserved immune or stromal cells) across conditions, while the separation supports the existence of condition-specific cell states or activation programs.

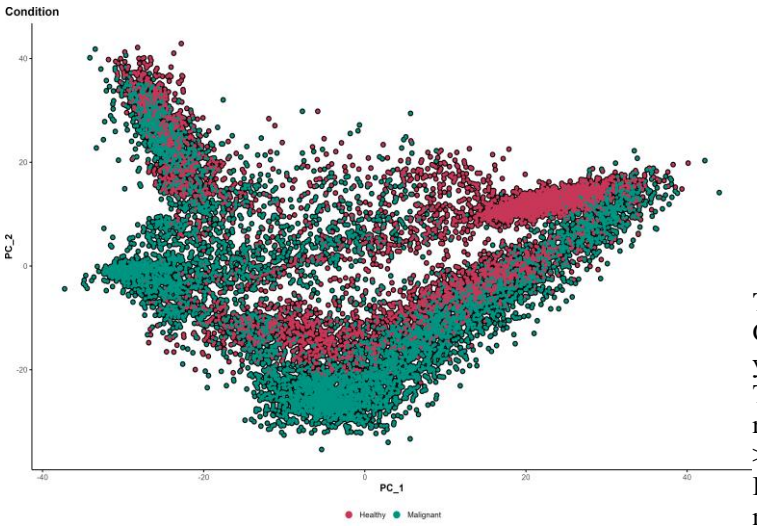


Figure 10; *Condition PCA*

b) Projection of QC Metrics onto PCA Space

To evaluate the spatial distribution of quality control (QC) metrics within the transcriptomic landscape, we projected library size (nCount_RNA), gene feature count (nFeature_RNA), and mitochondrial gene percentage (percent.mt) onto the first two principal components (PC1 and PC2). These metrics were visualized across the healthy and malignant samples (Figures 13,14,15), providing insight into how technical or biological factors influence transcriptomic structure.

Library Size (nCount_RNA) (Figure 11)
Cells are colored according to total UMI count, from light yellow (low) to black (high). In both samples, most cells exhibit moderate library sizes (10,000–30,000 UMIs).
The malignant sample shows broader variability, with a prominent cluster of cells exhibiting elevated library sizes (>30,000), especially in central and left portions of PC space. These regions may correspond to transcriptionally hyperactive or clonally expanded malignant populations.
The healthy sample also shows elevated library sizes in some localized clusters, possibly reflecting cell types like plasma cells or activated immune cells.

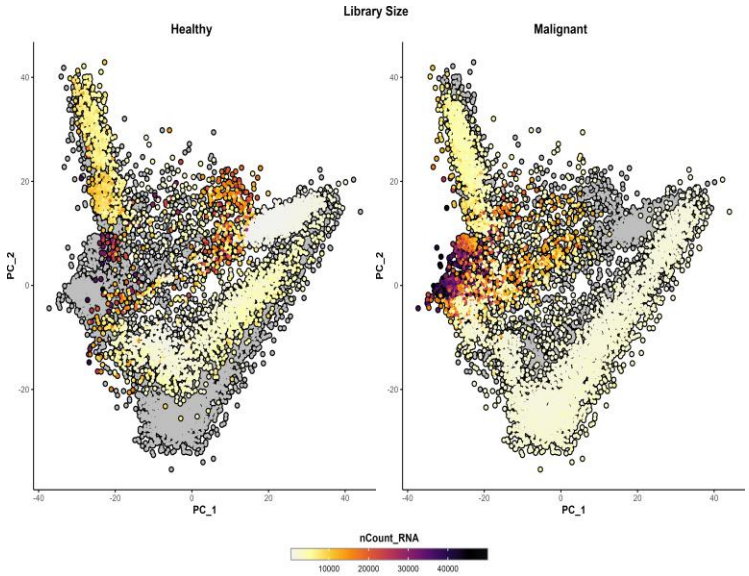


Figure 11; *Library size PCA*

Total Features (nFeature_RNA) (Figure 12)
Cells are colored by the number of detected genes, from light yellow (low feature count) to black (high feature count).
The malignant sample again displays a broader and higher range of gene detection, with several dense regions showing >4,000 genes per cell.
In contrast, the healthy sample has more compact and modestly distributed feature counts, primarily ranging from 1,000–3,000. These patterns confirm previous findings from histogram and violin plots, reinforcing that the malignant condition contains greater transcriptional complexity and heterogeneity.

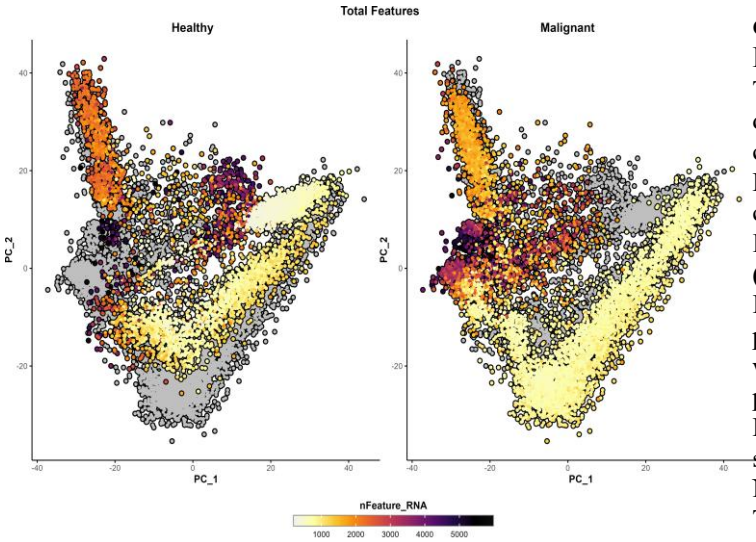


Figure 12; *Total features PCA*

Mitochondrial Gene Percentage (percent.mt) (Figure 13)
Cells are colored by the proportion of mitochondrial gene content, from yellow (0%) to black (>10%).
In both samples, most cells fall below 10%, indicating good cell quality overall.
However, the malignant sample shows more spatially clustered areas with elevated mitochondrial content (orange to dark purple/black), suggesting the presence of cells under metabolic stress or apoptosis.
These mitochondrial-high regions may reflect dying malignant cells, stressed immune cells, or cells in transition.

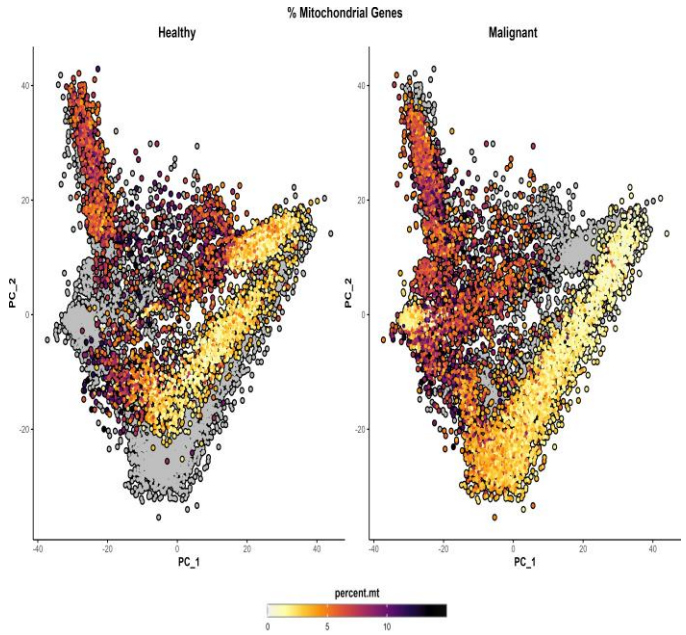


Figure 13; *Percentages mitochondrial genes PCA*

c) Principal Component Analysis After Cell Cycle Regression:
To eliminate cell cycle-induced transcriptional bias from downstream analyses, we regressed out the influence of cell cycle scores (G1, S, and G2/M phases). We then re-performed PCA and visualized the results to assess how the removal of cell cycle effects influenced global transcriptomic structure.
PCA Colored by Cell Cycle Phase (Post-Regression) (Figure 14):
In (Figure 14), cells are colored by their assigned cell cycle phase after regression. Unlike the pre-regression PCA plot, in which G2M and S-phase cells occupied distinct regions, the post-regression PCA shows:
Homogeneous mixing of G1, S, and G2M cells across PC space.
No discernible clusters driven by proliferative states.
This confirms that the cell cycle regression step effectively minimized the contribution of cycling gene programs to the principal components, allowing non-cell-cycle biological variation to dominate.

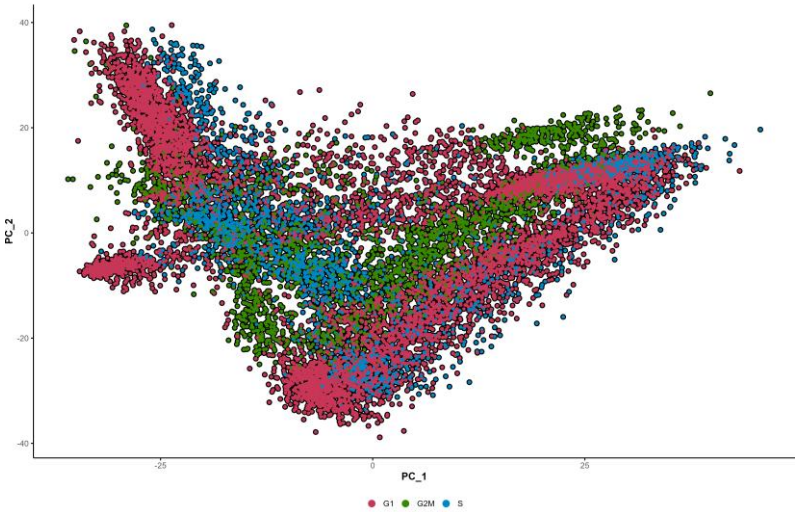


Figure 14; *Cell cycle phase PCA (After Cell cycle regression)*

PCA by Condition and Split View of PC4–PC5(Figure 15):
In a focused comparison of **PC4 vs PC5** (Figure 15), we visualized the full dataset colored by condition (left),
Healthy cells only (middle), and Malignant cells only (right).
This split-panel view reveals that:
PC4 and PC5 partially separate cells by condition, with malignant cells dominating certain projections.
Within-condition heterogeneity is also evident, especially in the malignant sample, which displays a broader spread in PC space.

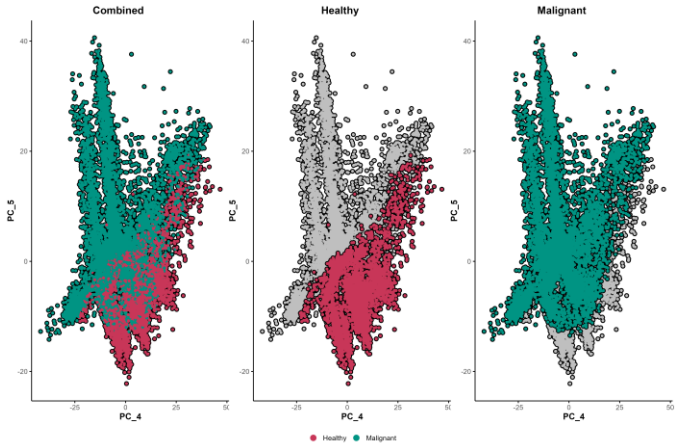


Figure 15; *Condition PCA (After Cell cycle regression)*

Elbow Plot for Principal Component Selection (Figure 16)

The elbow plot (Figure 16) depicts the standard deviation of each principal component (PC). It is used to determine the number of PCs that capture the majority of the variance for downstream clustering and visualization.

A distinct inflection point is observed around PC 15–20, after which the explained variance drops off sharply.

Based on this pattern, the first 20 PCs were retained for downstream analyses such as clustering and UMAP projection.

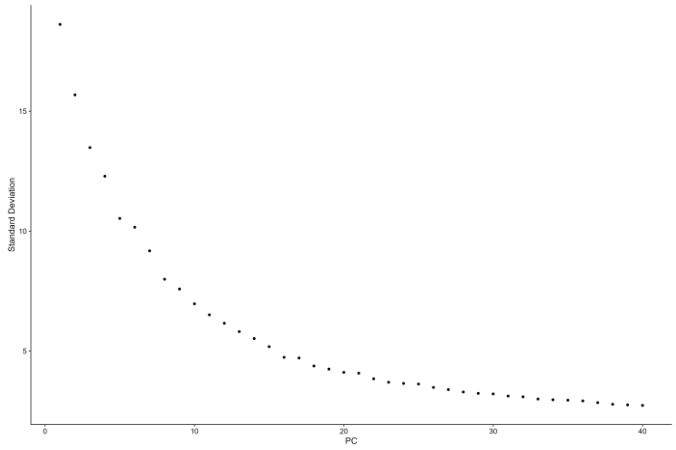


Figure 16; *PCA Elbow plot*

2) UMAP Visualizations Confirm Biological Separation

A Uniform Manifold Approximation and Projection (UMAP) was performed to visualize the global transcriptional landscape of single cells from both healthy and malignant samples.

a) Condition-Based UMAP Plot (Figure 17):

Each point represents a single cell projected onto two UMAP dimensions (UMAP_1 and UMAP_2).

Cells are colored by condition:

Red: Healthy

Green: Malignant

Observations:

The two conditions form distinct clusters, suggesting substantial transcriptional differences between healthy and malignant cells. Some overlap is observed between conditions in certain regions, which may reflect shared cellular states or transitional populations.

Malignant cells occupy separate regions in the UMAP space not populated by healthy cells, indicating condition-specific subpopulations and potentially malignant-specific transcriptional programs.

Healthy cells are more concentrated in the lower half and left arm of the UMAP, while malignant cells are more prevalent in the upper and right portions.

Interpretation:

The clear separation of clusters demonstrates successful dimensionality reduction and supports that the downstream clustering and differential expression analyses can distinguish between healthy and malignant states.

The presence of overlapping areas also indicates that some transcriptional programs may be shared, potentially corresponding to non-malignant cell types or microenvironmental influence.

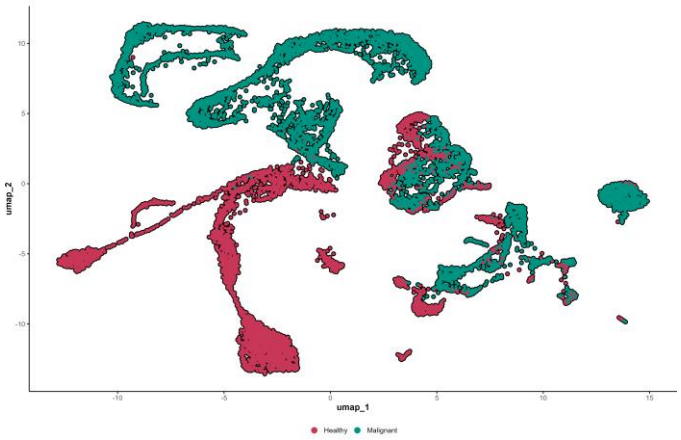


Figure17; *Condition UMAP*

b) UMAP Projection by Cell Cycle Phase (Figure18)

A UMAP dimensionality reduction was performed to visualize cells based on their transcriptional profiles, with cells colored by their predicted **cell cycle phase** using canonical markers.

Each dot represents a single cell plotted in UMAP space.

Cells are colored by their predicted phase of the cell cycle:

Red: G1 phase

Green: G2/M phase

Blue: S phase

Observations:

Cells from all three cell cycle phases are distributed throughout the UMAP space, with distinct but partially overlapping distributions.

The G1 phase (red) is the most abundant and widely distributed, dominating large areas of the UMAP.

S phase (blue) and G2/M phase (green) cells form discrete clusters, particularly in regions separate from the G1-dominated zones.

Specific regions show enrichment of actively cycling cells (S and G2/M), which may correspond to proliferative populations such as malignant cells or activated immune cells.

Interpretation:

The observed distribution confirms the presence of transcriptionally distinct cell states associated with different stages of the cell cycle.

The overlap of phases in some regions highlights the continuity of the transcriptional landscape across the cell cycle.

These results are critical for deciding whether **cell cycle regression** is needed prior to downstream analyses such as clustering or differential expression, particularly to avoid capturing variance driven solely by proliferative activity.

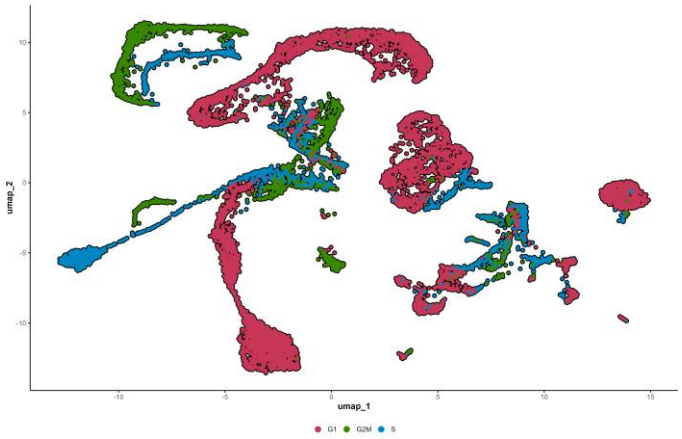


Figure18; *Cell Cycle Phase UMAP*

c) Differential Expression Analysis

The Wilcoxon-rank-sum test contrasted malignant against healthy cells at a $\log_2$-fold-change ($\log_2$FC) cutoff of ±0.25 and an FDR-adjusted p-value < 0.05.

Statistic	Value
Total significant genes	2061
Up-regulated in malignant	1017
Down-regulated in malignant	1044

Top up-regulated genes (malignant vs healthy):

IGLV3-1, IGLC2, JCHAIN, CAMP, DERL3, MZB1, CRISP3, OLR1, CHIT1, CXCL8 all show $\log_2$FC > 4 and adjusted $p < 1 \times 10^{-99}$.

Top down-regulated genes (malignant vs healthy)

CYP1B1, ALPL, CCR1, SIGLEC10, CD14, CREB5 present $\log_2$FC < -4.3 with similarly strong statistical support.

These figures are stored in write/tables/Markers.csv, which includes every DEG annotated as "Up" or "Down" for rapid filtering in downstream analyses.

Visual summaries

Volcano plot (Figure19)

The volcano plot (Fig.19) maps each of the 2061 DEGs by effect size (x-axis) and $-\log_{10}$ FDR (y-axis).

A clear bimodal distribution is evident, reflecting roughly equal numbers of up- and down-regulated genes.

Immunoglobulin-related transcripts such as IGLC2 (+7.6 $\log_2$FC), IGLV3-1 (+9.9 $\log_2$FC) and plasma-cell chaperone MZB1 (+5.3 $\log_2$FC) dominate the extreme right, highlighting the plasma-cell–like transcriptional program of malignant cells.

Innate-immune and myeloid markers including CD14, CCR1, and SIGLEC10 anchor the far-left quadrant, confirming their relative depletion in malignant samples.

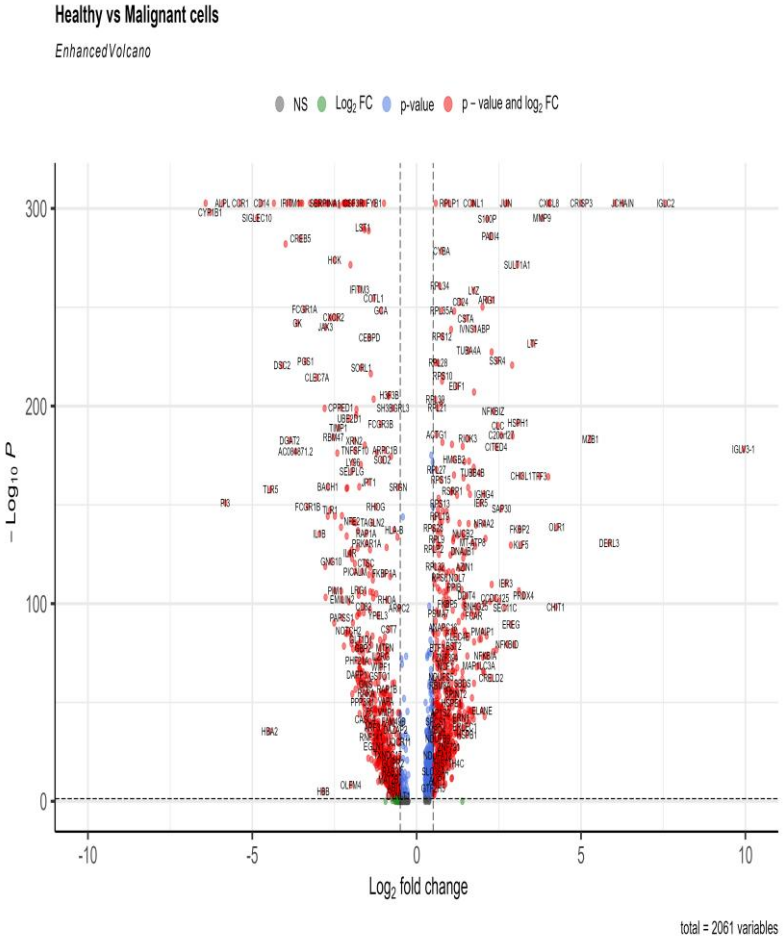


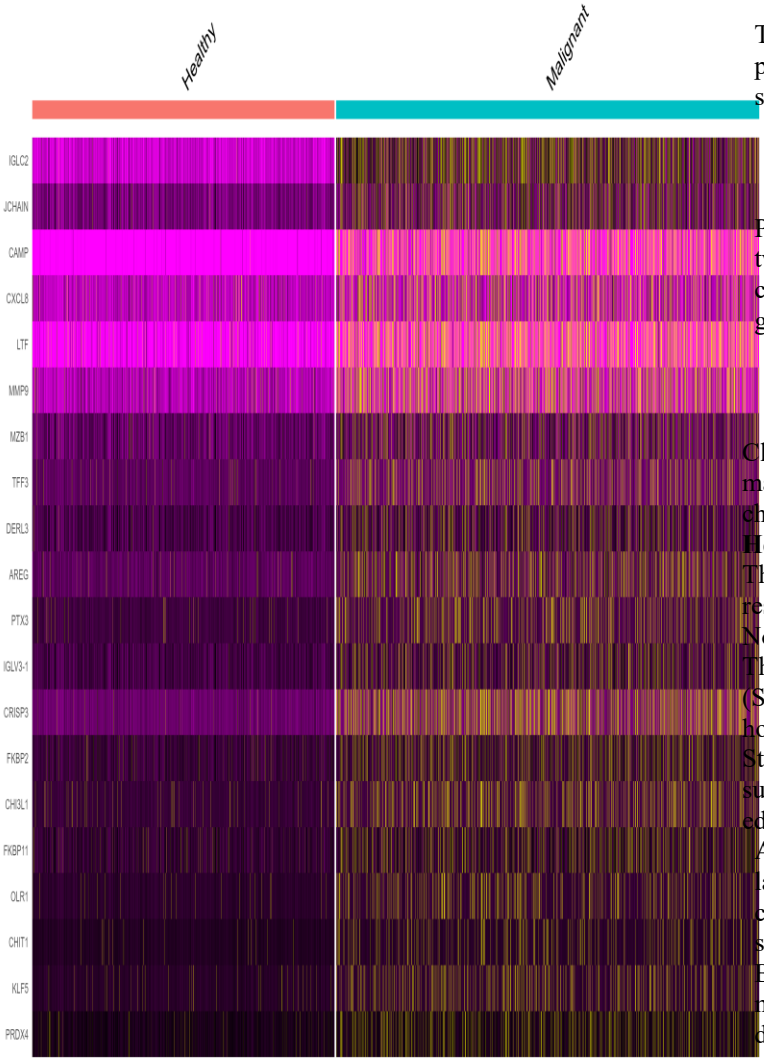
Figure19; *Volcano Plot*

Figure20; *Heatmap*

Heatmap

The condition heatmap (Fig.20) displays scaled expression of the 20 genes with the highest positive log_2FC.

- Healthy cells (left) show uniformly low or negative z-scores for these transcripts, while malignant cells (right) exhibit strong positive expression, producing a stark bicluster pattern that cleanly separates the two groups.
- The gene set is enriched for secretory and inflammatory mediators (e.g. CXCL8, PTX3, AREG) and endoplasmic-reticulum stress components (DERL3, FKBP2), consistent with an activated secretory phenotype in the malignant compartment.
- The clear segregation validates the statistical thresholds and underlines the robustness of the malignant transcriptional signature.



Biological interpretation

The DEG profile indicates that malignant cells:

- Up-regulate immunoglobulin light-chain transcripts and plasma-cell chaperones, consistent with expanded malignant plasma or B-cell–like populations.
- Over-express pro-inflammatory factors (CXCL8, MMP9, PTX3), which may remodel the bone-marrow micro-environment and recruit supportive stromal or immune cells.
- Down-regulate myeloid receptors (CD14, CCR1, TLR5) and inhibitory lectins (SIGLEC10), pointing to a loss of classical monocyte features in malignant clones.
- Exhibit decreased expression of hemoglobin genes (HBA2, HBB), reflecting the lower erythroid representation often observed in myeloma bone marrow.

Together, these results provide a quantitative foundation for pathway enrichment and cell–cell communication analyses in subsequent sections of the report.

d) Splitting the Combined Seurat Object into two Objects by condition

Prior to clustering evaluation, the Seurat object was split into two separate objects, one for healthy and one for malignant conditions. All downstream clustering trees and UMAPs were generated independently for each condition.

e) Clustering Tree Evaluation Supports Resolution 0.4

Clustering trees were generated for both the healthy and malignant datasets to examine how cell cluster assignments change across different resolutions.

Healthy Sample Clustering tree: (Figure 21)

The left side of the clustering tree represents increasing resolution from top to bottom.

Node size corresponds to the number of cells in that cluster.

The node color represents the clustering resolution (SCT_snn_res.), while edge thickness and directionality show how clusters split or merge across resolutions.

Stability is visualized with purple edges. Lighter purple suggests less consistent cluster membership, while yellow edges indicate more stability.

At lower resolutions (e.g. 0.1–0.4), the clusters are fewer and larger. As the resolution increases (e.g. ≥ 0.6), more distinct clusters emerge, which likely correspond to finer subpopulations.

Beyond resolution ~0.6, many splits are observed without major increases in biological interpretability, suggesting diminishing returns.

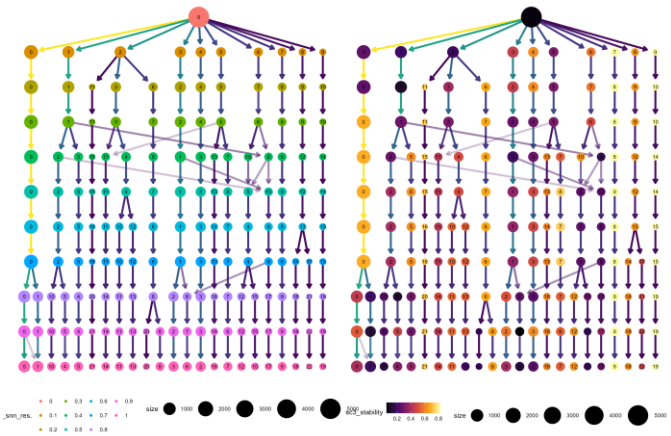


Figure21; *Healthy clustering trees*

Malignant Sample Clustering tree (Figure 22):
Follows the same design.
A similar trend of initial stability followed by increasing granularity is observed.
However, the malignant sample shows greater instability at higher resolutions (thicker purple branches), reflecting the heterogeneity and potential plasticity of malignant cells.
The instability is most notable between resolutions 0.4 and 0.9, implying high sensitivity of cluster assignment to resolution in malignant cells.

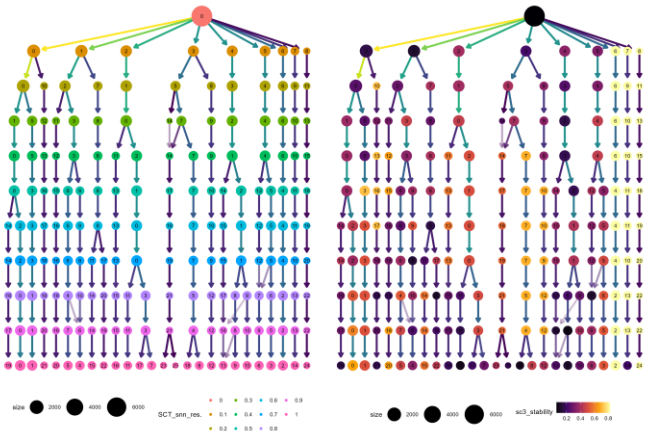


Figure22; *Malignant clustering tree*

f) UMAP Projection of Seurat Clusters

UMAP plots were used to visualize the final clustering of both datasets based on the selected resolution (0.4).
Healthy UMAP (Resolution 0.4) (Figure23)
The cells are grouped into 16 distinct clusters, labeled 0 through 15.
Clusters are generally well separated with clear spatial boundaries, suggesting that cell types are transcriptionally distinct.
No major overlaps or ambiguous regions are observed, confirming a high-quality clustering result.

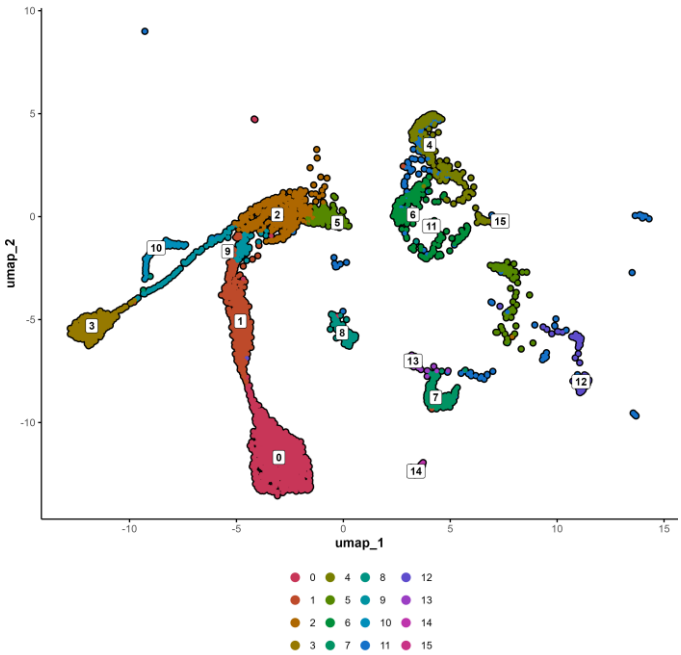


Figure23; *healthy Seurat Clusters UMAP*

Malignant UMAP (Resolution 0.4) (Figure 24)
Also shows 16 clusters, but with more overlap between neighboring groups, consistent with the instability seen in the Clustering tree.
Clusters 0, 1, and 2 dominate large regions of the UMAP, suggesting either:
Expansion of certain malignant subtypes.
Reduced transcriptional distinctiveness within the malignant compartment.
Smaller, more discrete clusters also exist (e.g. clusters 6, 10, 15), possibly representing rare malignant subpopulations or contaminating healthy cells.

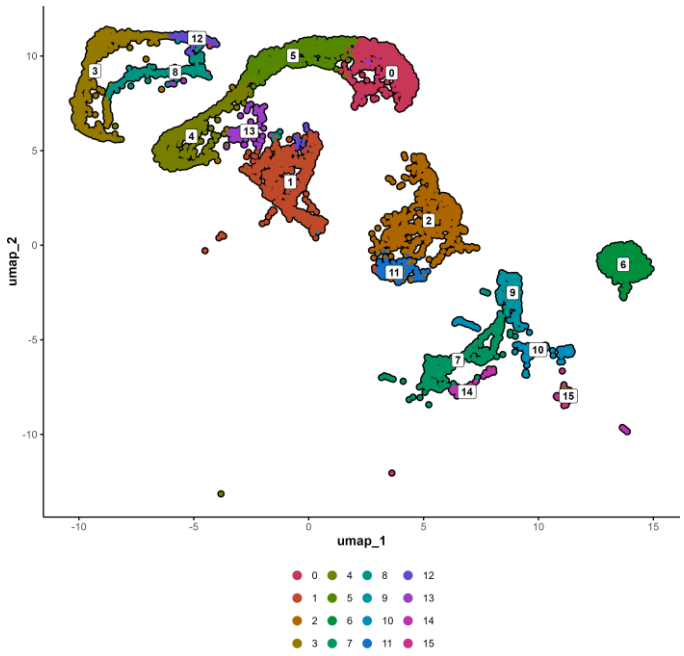


Figure24; *malignant Seurat Clusters UMAP*

g) Feature Expression Analysis (Figure25)

To investigate marker gene expression across different malignant cell clusters, UMAP feature plots were generated for six genes: DEFA3, IGLC2, JCHAIN, CAMP, MPO, and PRTN3. Each gene's expression is visualized across the UMAP embedding of the malignant sample, with intensity corresponding to log-normalized expression levels. These genes were automatically selected as the top six highly variable features using Seurat's VariableFeatures() function.

DEFA3

DEFA3 (defensin alpha 3), a granulocyte marker, showed strong and spatially restricted expression in a specific cluster located in the lower-central region of the UMAP. This suggests DEFA3 marks a subset of cells likely corresponding to mature neutrophils or granulocytic lineage populations within the malignant sample.

IGLC2

IGLC2 (immunoglobulin lambda constant 2) exhibited broad but lower-intensity expression, particularly across multiple peripheral clusters. This diffuse expression pattern is consistent with its role in B cells or plasma cells, which may be variably represented within malignant compartments.

JCHAIN

JCHAIN (joining chain of multimeric IgA and IgM) showed low to moderate expression in scattered peripheral regions, with no dominant peak cluster. The expression pattern suggests heterogeneous plasma cell distribution with lower enrichment in any single malignant subpopulation.

CAMP

CAMP (cathelicidin antimicrobial peptide) displayed a highly similar pattern to DEFA3, with intense expression in the same central cluster, reinforcing its association with the granulocytic lineage. Co-expression with DEFA3 indicates a robust neutrophil signature in this region of the UMAP.

MPO

MPO (myeloperoxidase), another granulocyte-associated gene, showed high expression in the DEFA3/CAMP region, further validating the neutrophil identity of this cluster.

Unlike DEFA3 and CAMP, MPO showed a slightly broader distribution, possibly marking earlier granulocyte precursors in adjacent clusters.

PRTN3

PRTN3 (proteinase 3) expression was localized and intense, again in the central cluster region shared with DEFA3, CAMP, and MPO. The co-expression of PRTN3 and MPO is characteristic of neutrophil-like malignancies and may serve as a functional or diagnostic marker in this context.

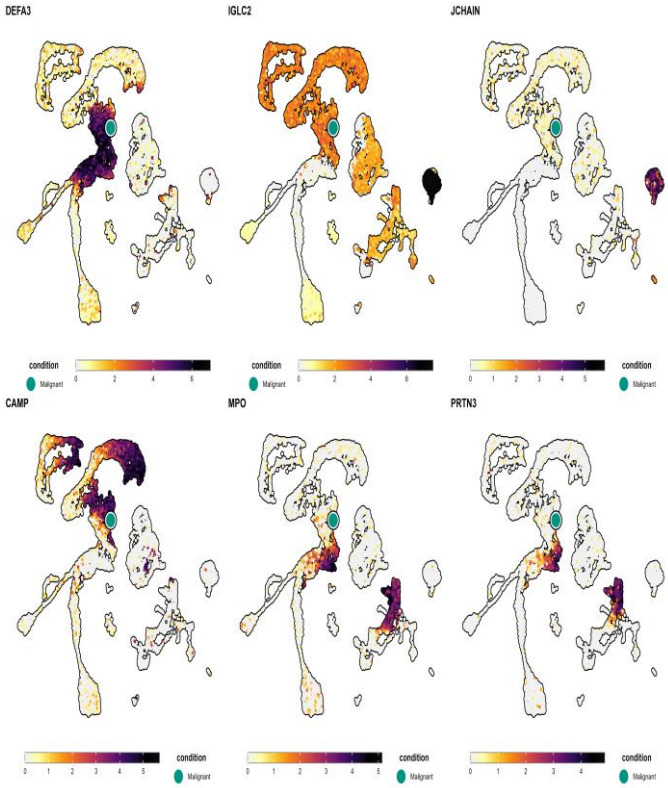


Figure25; *Features_FeaturePlots*

5) Cell Type Annotation

To elucidate the cellular composition of the bone marrow microenvironment in both healthy and malignant conditions, we performed cell type annotation using the SingleR framework. Two complementary strategies were employed:

1. Cluster-based annotation, where cells were assigned a label based on their cluster identity.
2. Cell-based annotation, providing a finer resolution by annotating each cell individually.

For reference, we used the Human Primary Cell Atlas provided by the celldex package, which contains transcriptomic profiles of a broad array of primary human cell types.

a) Healthy Bone Marrow

Cluster-based Annotation:
Cluster-level annotation assigned specific immune and progenitor cell types to each Seurat cluster in the healthy dataset. The following major cell types were identified:
Hematopoietic stem/progenitor cells: HSC-G-CSF, CMP, GMP, MEP
Immune cells: T cells, B cells, NK cells, Monocytes, Neutrophils
Erythroid lineage: Erythroblasts, Platelets
B-cell lineage intermediates: Pro-B_cell_CD34+, Pre-B_cell_CD34-
These annotations were largely consistent with canonical lineage expectations in healthy bone marrow and aligned well with cluster-specific marker gene expression (from FindAllMarkers()).
Cell-based Annotation:
At the single-cell level, annotations preserved the diversity of hematopoietic and immune cell types observed at the cluster level, while revealing finer heterogeneity within certain clusters. Importantly, rare populations such as Pro-B cells and Pre-B cells were retained and not masked by cluster averaging. Cells without confident matches were excluded from downstream comparisons.

b) Malignant Bone Marrow

Cluster-based Annotation:
The malignant dataset displayed a broadly similar cellular landscape to the healthy reference, including:
Myeloid cells: Monocytes, Neutrophils, Pro-Myelocytes, Myelocytes
Lymphoid lineage: B cells, T cells, NK cells
Progenitor and stem cells: GMP, MEP, CMP, HSC-G-CSF
Erythroid lineage: Erythroblasts, Platelets
Bone marrow stroma (BM)
This indicates that the malignant microenvironment still maintains a core set of hematopoietic populations, albeit potentially with altered abundance or function.
Cell-based Annotation:
Single-cell resolution again confirmed the presence of all major hematopoietic lineages, with some clusters displaying higher heterogeneity. Importantly, some **non-hematopoietic or ambiguous cell types** were filtered out to ensure focus on comparable populations.

C) Shared Cell Type Filtering

To ensure meaningful differential analyses and intercellular communication studies, we retained only cell types shared between healthy and malignant datasets. This resulted in 16 consistent cell types across both conditions, including:
Stem/progenitor cells: HSC-G-CSF, CMP, GMP, MEP
Immune cells: T_cells, B_cell, NK_cell, Monocyte, Neutrophils
B-cell intermediates: Pro-B_cell_CD34+, Pre-B_cell_CD34-
Erythroid lineage: Erythroblast, Platelets
Others: BM, Myelocyte, Pro-Myelocyte
This filtering step is crucial for avoiding bias in downstream comparative analyses and ensures that observed differences are attributable to disease-related expression changes, not the presence or absence of specific cell types.

6) Intercellular Communication Analysis with CellChat

To systematically assess how cellular communication is altered in disease, we utilized CellChat to infer ligand-receptor (LR) signaling networks from both Healthy and Malignant bone marrow datasets**.**

a) Network Construction and Filtering

After initializing chat_healthy and chat_malignant objects with Seurat data, CellChat's pipeline was used to:
Identify overexpressed genes and ligand-receptor interactions, compute communication probabilities using the triMean method and filter out interactions from underrepresented cell groups (e.g., MEP, Pro-B_cell_CD34+ in healthy; BM, Platelets, Pro-B_cell_CD34+ in malignant).
Key findings:

- 454 and 502 LR pairs were used in healthy and malignant samples, respectively.
- 20.8% of interactions were removed from the healthy dataset; 27.8% from the malignant due to cell group filtering.

b) Inferred Signaling Pathways

The analysis revealed a total of 36 signaling pathways in healthy samples and 22 in malignant samples. While some pathways were shared, key differences were evident. common pathways:
Both conditions shared core immunomodulatory and inflammatory pathways such as:

- MIF, GALECTIN, ANNEXIN
- MHC-I, RESISTIN, CXCL, CD99, CLEC

Healthy-specific Pathways:

- APP, CTSG, SELL, IL16, SEMA4, NEGR, LCK, CD23, PECAM2, BTLA, etc.
- Suggest a richer and more diverse communication network, indicative of a well-functioning and regulated bone marrow niche.

Malignant-specific Pathways:

- CysLTs, THBS (Thrombospondin), and a few immune-evasive or pro-inflammatory signals not active in healthy tissue.
- Suggest disruption in homeostasis, increased inflammation, and possible immune escape mechanisms.

c) Comparative Interaction Networks

Overall Cell-Cell Communication Activity (Figure 26)

To evaluate global cell-cell communication patterns between healthy and malignant bone marrow samples, we computed two summary metrics using CellChat: (1) the total number of inferred interactions and (2) the overall interaction strength (cumulative communication probability across all significant signaling pathways).

Number of Inferred Interactions

The total number of significant ligand-receptor interactions was notably higher in the healthy condition, with 872

interactions**,** compared to only 602 interactions in the malignant condition. This reduction of 31% in the malignant sample suggests a substantial loss of signaling diversity or engagement among cellular compartments in the diseased state.

Overall Interaction Strength

Similarly, the cumulative interaction strength was higher in the healthy condition, measured at 20.142**,** versus only 11.876 in the malignant condition. This represents a 41% reduction in global signaling intensity**,** indicating that even among retained interactions, the magnitude or confidence of communication is attenuated in malignancy.

Interpretation

Together, these findings indicate that malignant transformation is associated with both a qualitative and quantitative decline in intercellular signaling. The observed reductions suggest a global disruption of tissue homeostasis, likely contributing to tumor progression, immune evasion, or stromal reprogramming in the malignant bone marrow environment.

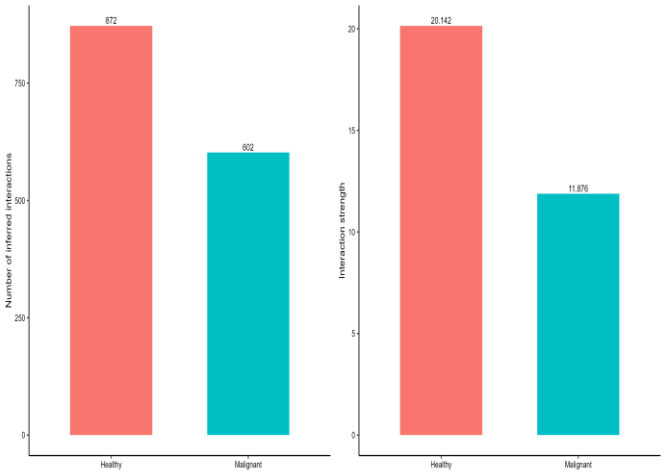


Figure26; *total interaction counts & weights*

Differential Interaction Networks

To pinpoint how specific cell-cell interactions change between healthy and malignant bone marrow, we visualised differential interaction networks with CellChat's netVisual_diffInteraction().

Differential Number of Interactions (Figure27):

Each edge indicates a change in the count of significant ligand–receptor pairs from healthy to malignant. Blue edges represent interactions that decrease in malignancy, whereas red edges denote interactions that increase. The dominance of blue edges highlights a widespread loss of signalling diversity, particularly among neutrophils, NK cells, and monocytes.

Differential Interaction Strength (Figure28):

Here, edge width and color encode the change in cumulative communication probability (weight) for each cell-type pair. Again, blue edges prevail, confirming that most retained interactions are **weaker** in the malignant micro-environment. A few red edges (e.g. GMP → HSC-G-CSF and Erythroblast loops) reveal pathways that become locally stronger and may favor malignant growth.

Interpretation:

Together, Figures 27 and 28 show that malignant transformation leads to a broad **contraction of both the number and strength of intercellular communications**, with only selective gains in specialised niches. This rewiring likely contributes to immune evasion, impaired tissue homeostasis, and the emergence of tumour-supportive signalling hubs within the bone-marrow micro-environment.

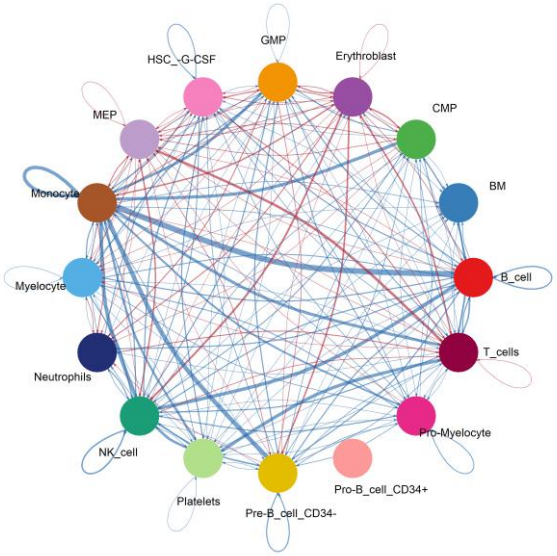


Figure27; *diff. interaction counts*

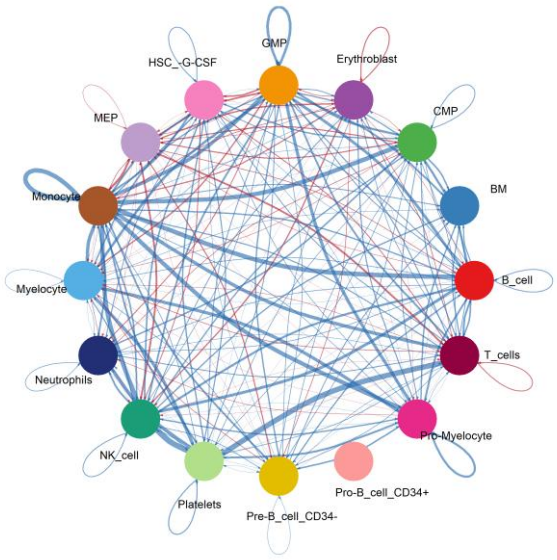


Figure28; *diff. interaction weights*

d) Cell-Cell Communication Heatmaps

To assess how disease state alters intercellular signal patterns across specific cell types, we generated heatmaps differential interaction counts and interaction streng between healthy and malignant samples.

Differential Number of Interactions

The left heatmap (Figure29) displays the difference in number of inferred interactions for each sender-receiver p Red indicates a higher interaction count in malignant, wl blue indicates a higher interaction count in healthy.

Erythroblasts, GMPs, and MEP**s** in the malignant condit showed a notably higher number of outgoing signals (rov toward various cell types, especially toward other myeloid a progenitor cells.

Neutrophils and Myelocytes show an overall decrease interaction numbers in the malignant condition (blue tiles), indicating a loss of myeloid communication relative to healthy marrow.

The strongest reductions in signaling occurred between:

- **Erythroblasts → CMP, GMP, and MEP**
- **MEPs → multiple progenitor and differentiated myeloid cells**
- **Differential Interaction Strength**

The right heatmap (Figure29) quantifies the difference in communication strength, again highlighting red for higher levels in malignant and blue for higher in healthy.

- Most cell pairs actually display a global decrease in interaction strength in the malignant condition, with only localized gains in selected progenitor–progenitor edges.
- Notably, erythroblasts, monocytes, and MEPs exhibit stronger outgoing communication in the malignant samples, whereas their signaling is dampened in healthy marrow.

Interpretation

The heatmap comparison indicates that malignant transformation broadly rewires intercellular communication, particularly in:

- Hematopoietic progenitors (erythroblasts, GMPs, MEPs)
- Immune cells (T cells, B cells)

This rewiring of signaling networks likely reflects a shift in tissue function, favoring pro-tumor or immune-evasive states in the malignant bone marrow microenvironment.

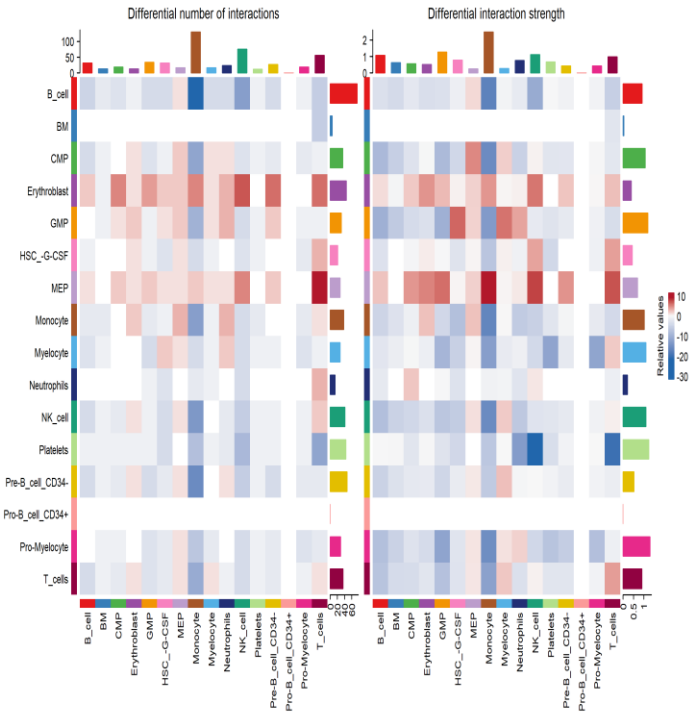


Figure29; *heatmaps interaction counts, weights*

e) Cell–Cell Communication Network Analysis (Circle Plots)

To comprehensively compare intercellular communication landscapes between healthy and malignant bone marrow, we visualized two complementary aspects:

- **Interaction Strength**
- **Number of Inferred Interactions**

Each node represents a cell type, and edges represent inferred communication events based on known ligand–receptor interactions. Thicker and more saturated edges denote stronger or more numerous interactions.

12. Number of Interactions Comparison (Figure31):

Left Panel: Healthy Bone Marrow
Right Panel: Malignant Bone Marrow

The number of inferred interactions is also significantly reduced in the malignant setting.

Healthy samples display densely connected networks, with broad communication among all major hematopoietic populations. Especially, CMP, GMP, Monocytes, and Platelets form dense interaction hubs in the healthy tissue, suggesting key roles in normal homeostasis.

In malignancy, the network is sparser, with several cell t such as BM, Platelets, and NK_cells exhibiting limited o connectivity. This sparsity may reflect impa differentiation, altered receptor–ligand dynamics, or cl dominance of malignant populations suppressing normal crosstalk.

These circle plots confirm that malignancy substant reduces both the strength and the number of intercell communications, affecting multiple hematopoietic immune compartments. The data point toward a disru signaling environment in malignant bone marrow, which contribute to tumor progression and immune evasion.

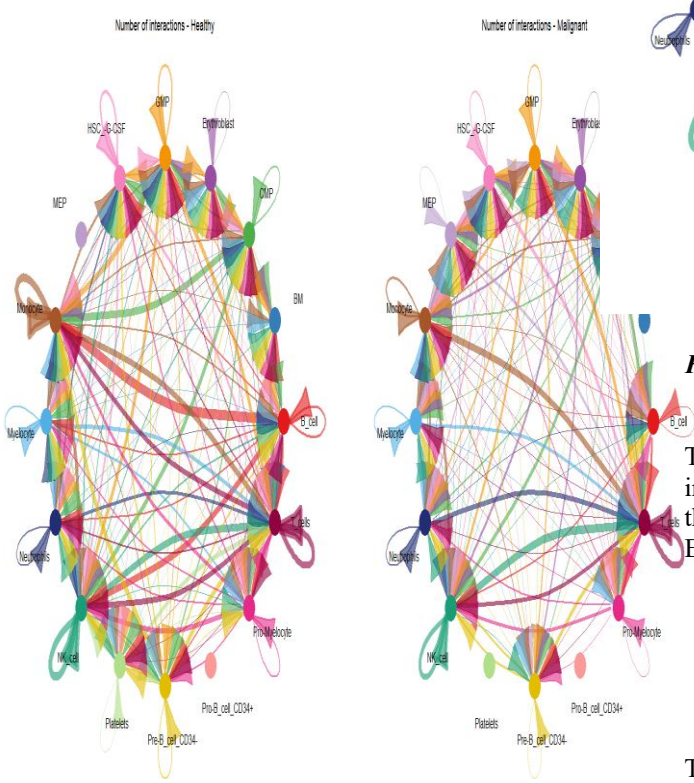


Figure30; *Circle interactions counts*

1. Interaction Strength Comparison (Figure30):

Left Panel: Healthy Bone Marrow
Right Panel: Malignant Bone Marrow

The healthy network shows more intense and numerous strong interactions, reflected by the thicker and more saturated ribbons between multiple cell types.

In particular, Monocytes, Neutrophils, HSC-G-CSF, GMP, and T_cells display robust outgoing and incoming interactions, indicating an active, highly coordinated signaling network.

In contrast, the malignant condition exhibits fewer and weaker interactions, shown by the relatively thinner and less saturated connections across the network.

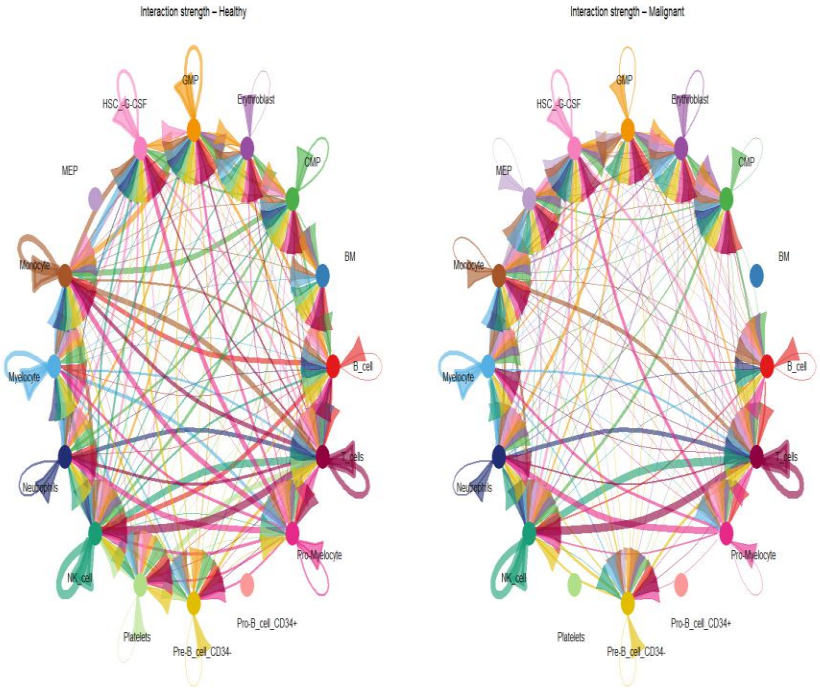


Figure31; *Circle interactions weights*

f) Signaling Role Scatter Plots (Figure32)

This plot (Figure32) compares the incoming vs. outgoing interaction strengths for each cell type, giving insight into their functional roles in intercellular communication networks. Each point represents a specific cell type, with:

- **X-axis:** Outgoing interaction strength (how actively a cell type sends signals)
- **Y-axis:** Incoming interaction strength (how much a cell type receives signals)
- **Point size:** Number of interactions (Count), representing the communication volume

The plots are split into two panels:

- **Left:** Healthy bone marrow
- **Right:** Malignant bone marrow

Findings:

Overall Communication Landscape

In the healthy condition, most cell types show balanced and distributed signaling, with a number of cells exhibiting both strong incoming and outgoing signals (e.g., T_cells, NK_cells, Monocytes).

In the malignant condition, the majority of cell types show a collapse in both incoming and outgoing communication strength, shifting toward the bottom-left quadrant, reflecting reduced communication.

Key Sender and Receiver Cells:

Cell Type	Healthy State	Malignant State
T_cells	High outgoing and incoming strength – active sender and receiver	Still high, though s communicatively a
NK_cells	Similar to T_cells, active in both roles	Maintains signalin
Monocytes	Strong sender and receiver	Still retains moder
Neutrophils	High incoming, low outgoing (primarily receivers)	Similar pattern ob
HSC_G-CSF	Strong receiver	Moderate incomin

The scatter plot (Figure32) highlights a global suppressi signaling activity across nearly all cell types in the mali bone marrow.
While T_cells, NK_cells, and Monocytes retain nc signaling roles under malignancy, most other hematop and progenitor populations are functionally silenced, v may reflect tumor-driven disruption of normal interce coordination.
These shifts underscore how malignancy remodels communication landscape, potentially fostering im evasion and blocking differentiation cues.

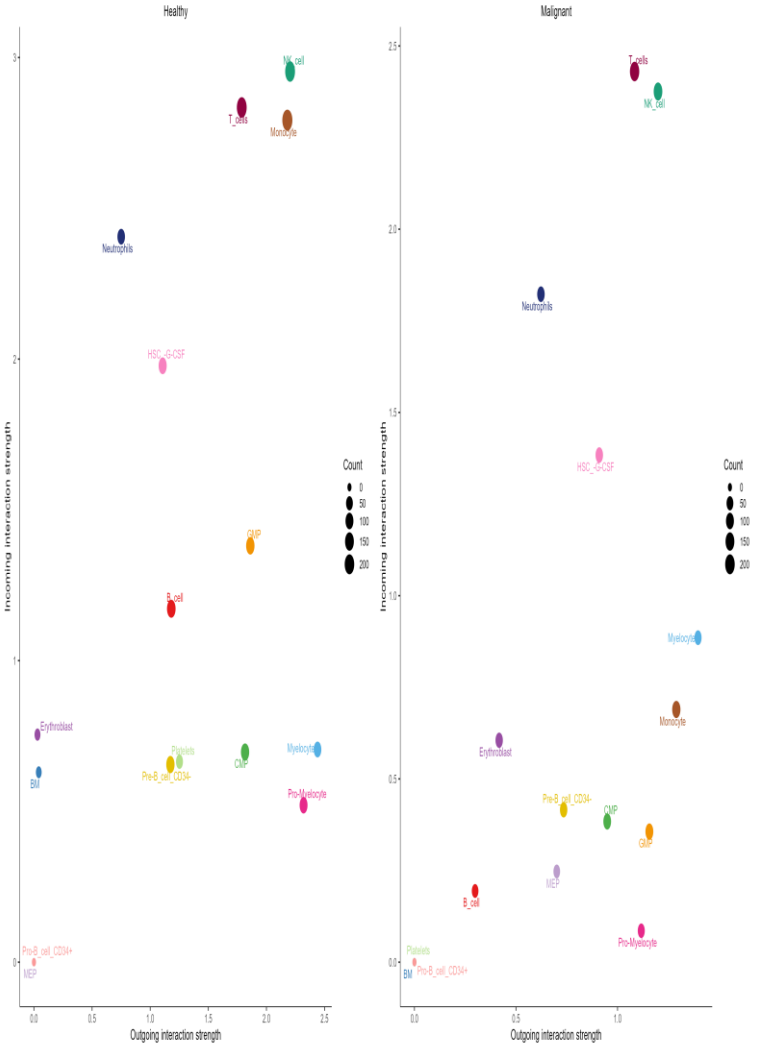


Figure32; *signaling role scatter*

g) Pathway-Specific Insights: MIF, GALECTIN, ANNEXIN

Why were the MIF, GALECTIN, and ANNEXIN pathways chosen?

The MIF, GALECTIN, and ANNEXIN pathways were specifically selected from the CellChat analysis due to their well-established roles in immunomodulation and suppression within the bone marrow microenvironment of multiple myeloma (MM). These pathways align directly with the biological context and composition of the dataset under study (GSE176131).

1. MIF Pathway
Macrophage Migration Inhibitory Factor (MIF) is frequently elevated in MM bone marrow. Through its interactions with the receptor CD74, either complexed with CXCR4 or CD44, MIF signaling promotes immunosuppressive polarization of macrophages toward an M2-like phenotype and has been implicated in driving exhaustion of infiltrating $CD8^+$ T cells. Given that the analyzed dataset contains diverse hematopoietic lineages, including subsets of T cells and myeloid cells expressing both MIF and its receptors, this pathway provides critical insights into the immunosuppressive signals encountered by immune cells within the MM microenvironment.

2. GALECTIN Pathway
Galectin-9 (LGALS9) is similarly elevated in the MM bone marrow and signals via multiple key receptors, including CD44, CD45, TIM-3 (HAVCR2), and the redox-associated receptor P4HB. Galectin-9 binding to these receptors is involved in crucial immunosuppressive processes, including stabilization and expansion of regulatory T cells (Tregs), polarization of macrophages toward an M2 phenotype, and functional exhaustion of cytotoxic lymphocytes. The CellChat analysis of the dataset revealed significant Galectin-9-driven communications across multiple immune populations, underscoring the biological relevance of Galectin-9-mediated immunosuppressive interactions in the studied MM microenvironment.

3. ANNEXIN Pathway
Annexin A1 (ANXA1) signals primarily through Formyl Peptide Receptors (FPR1 and FPR2) and is a well-documented regulator of inflammation resolution and immune suppression. ANNEXIN-FPR interactions have been shown to attenuate cytotoxic T cell and NK cell activity, enhance regulatory T cell functions, and promote an immunosuppressive myeloid environment. Within the analyzed dataset (GSE176131), CellChat detected substantial ANNEXIN-FPR signaling interactions among immune cell populations, illustrating critical immunosuppressive mechanisms relevant to the immune dysregulation seen in MM.

Bubble Plots:

Bubble plots comparing healthy and malignant samples showed:

MIF pathway (Figure33):
The MIF signaling pathway includes interactions of MIF with CD74 in complex with CXCR4 or CD44.
Both MIF–CD74+CXCR4 and MIF–CD74+CD44 complexes show moderate to high communication probabilities across a broad range of cell types.
Interaction probabilities are mostly lower in malignant cell pairs than in their healthy counterparts: of the interactions significant in either sample, 103 are stronger in the healthy and 31 in the malignant condition.
In the malignant sample, MIF is sent mainly by MEPs, GMPs, and erythroblasts and received mainly by monocytes, GMPs, NK cells, and pre-B cells.
These patterns indicate that MIF signaling is reduced in malignant bone marrow, in line with the overall decrease in cell-cell communication.

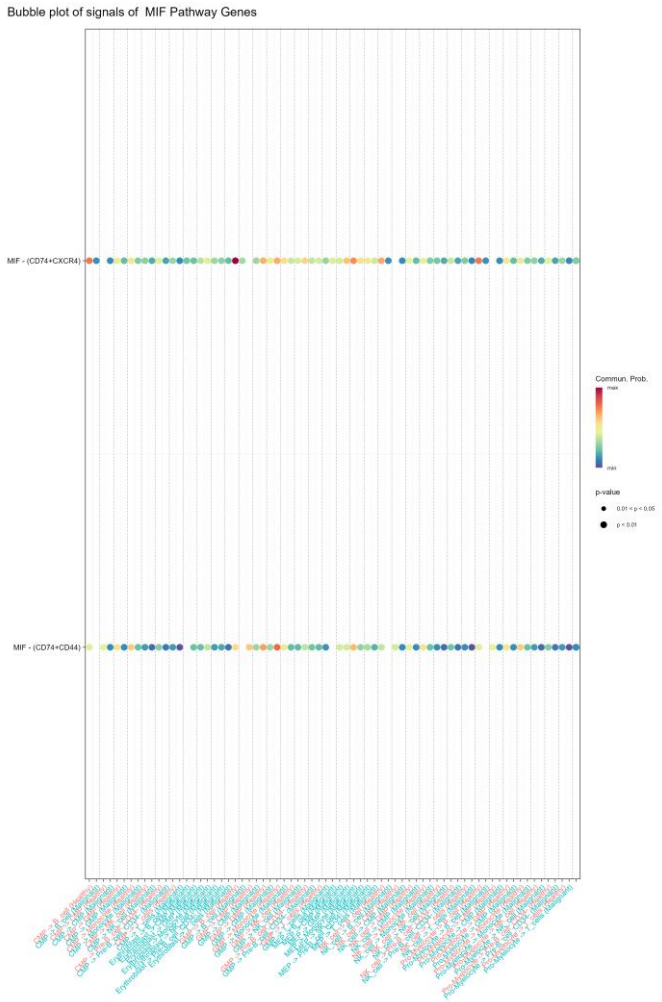


Figure33; *Bubble plot of MIF pathways*

GALECTIN Pathway (Figure34):

The GALECTIN signaling pathway is primarily represented by interactions involving LGALS9 as the ligand and several receptors including CD44, CD45, HAVCR2, and P4HB.

LGALS9-CD44 and LGALS9-CD45 interactions display the most widespread communication across cell pairs.

More LGALS9 interactions reach statistical significance only in the malignant sample (55, compared with 36 only in the healthy sample), mostly sent by monocytes, but their communication probabilities are slightly lower (median malignant-to-healthy ratio 0.85).

LGALS9-P4HB and LGALS9-HAVCR2 show weaker and more sporadic interaction probabilities, and LGALS9-HAVCR2 is not detected in the malignant sample.

The pattern suggests that galectin signaling spreads across more cell pairs in malignancy without becoming stronger; a contribution to immune dysfunction in the tumor microenvironment remains a hypothesis.

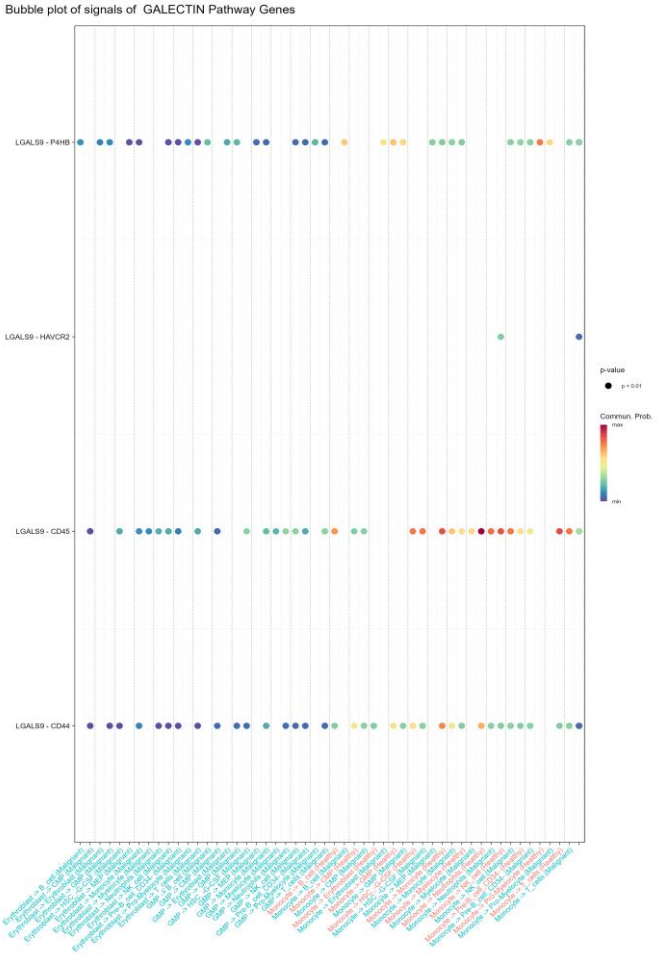


Figure34; *Bubble plot of Galectin pathways*

ANNEXIN Pathway (Figure35)

The ANNEXIN signaling pathway is represented here by ANXA1 ligands interacting with FPR1 and FPR2 receptors.

ANXA1-FPR1 interactions are ubiquitous and display the highest communication probabilities, but most cell pairs significant in both samples are weaker in malignancy (median ratio 0.31).

Statistically significant interactions (larger dots) are especially observed in granulocytic and monocytic lineage cells, consistent with FPR1's expression in innate immune cells.

ANXA1-FPR2 interactions are present but with lower communication probabilities and reduced statistical significance.



Figure35; *Bubble plot of Annexin pathways*

Hierarchy Plots:

Signaling Pathway Network Results (Circle Plots)

1. GALECTIN Signaling Network

- **Healthy Condition:**

Monocytes are the central signaling hubs, exhibiting extensive outgoing and incoming communication with multiple cell types including B cells, T cells, NK cells, neutrophils, and myelocytes.

The communication is broad and strong, with thick edges indicating high interaction strength.
Most interactions originate from monocytes, suggesting a dominant regulatory role.

- **Malignant Condition:**

While monocytes remain central, the overall number and strength of interactions appear reduced.
Interactions are more evenly distributed compared to the healthy state, and some connections are diminished in strength.
The monocyte-to-T cell and monocyte-to-B cell interactions remain relatively strong, but other links, such as with neutrophils and NK cells, are less pronounced.
Conclusion: GALECTIN signaling is heavily monocyte-driven in the healthy condition but becomes more fragmented and reduced in malignancy.

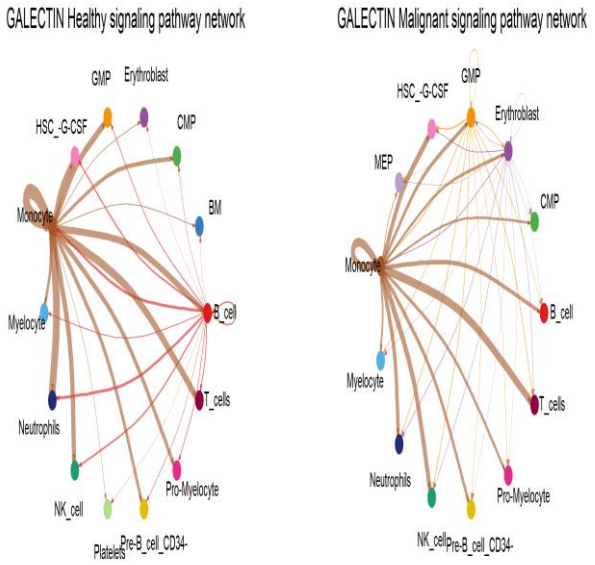


Figure36; *Hierarchy Galectin pathway*

2. ANNEXIN Signaling Network

- **Healthy Condition:**

A highly interconnected network with robust bidirectional signaling.
Neutrophils, monocytes, and HSC-G-CSF act as major signaling hubs.
Dense intercellular communication is observed, particularly from neutrophils to multiple targets including NK cells, CMP, and T cells.

- **Malignant Condition:**

The overall pattern of interaction is preserved, but a few edges show increased signal intensity, particularly from HSC-G-CSF and myelocytes.
Neutrophils maintain strong outgoing signals, especially towards HSC-G-CSF, myelocytes, and NK cells.
The pathway becomes slightly more centralized around specific nodes like HSC-G-CSF, suggesting a shift in regulation.
Conclusion: ANNEXIN signaling remains active in both conditions but becomes more focused in malignancy, with HSC-G-CSF emerging as a stronger hub.

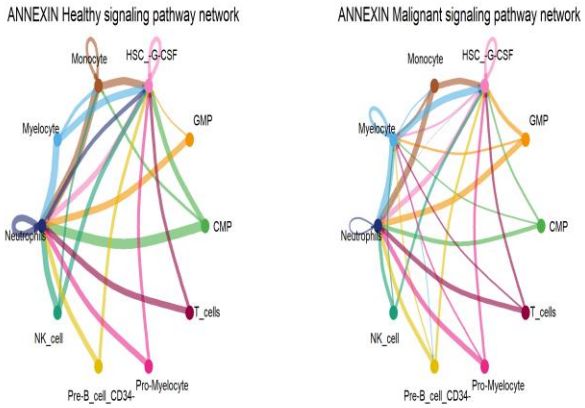


Figure37; *Hierarchy Annexin pathway*

3. MIF Signaling Network

- **Healthy Condition:**

The network shows dense signaling, with GMP and NK cells as major signal senders and receivers.
Broad communication exists between immune and progenitor populations (e.g., GMP, CMP, B cells, T cells, myelocytes).
Strong bidirectional interactions involving monocytes, NK cells, and CMP highlight a balanced signaling landscape.

- **Malignant Condition:**

The structure is altered, with a noticeable reduction in outgoing signaling from NK cells and monocytes.
GMP becomes a more prominent signaling hub, particularly in autocrine signaling (loop back to itself).
Some interactions, like those between monocytes and erythroblasts or between GMP and T cells, appear weaker or absent compared to the healthy state.

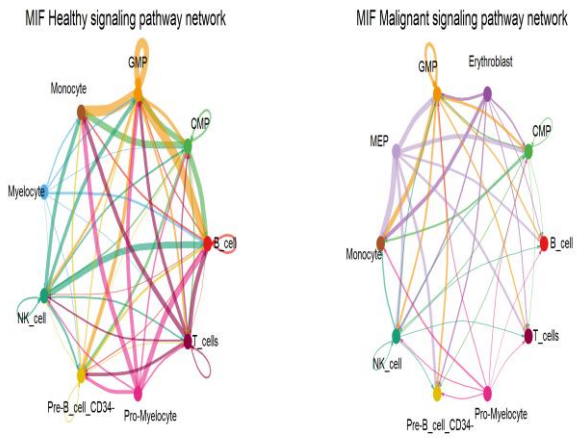


Figure38; *Hierarchy MIF pathway*

Violin Plots - Pathway Gene Expression Profiles

1.GALECTIN Pathway Genes (Figure39)

LGALS9 and HAVCR2 (TIM-3) are expressed mainly in Monocytes and NK cells, with lower expression in malignant samples, except for a slight increase in T cells.

CD44 shows consistent expression across both conditions in multiple cell types, particularly T cells and GMP.

P4HB is expressed in Pro-B cells and Pro-Myelocytes, with similar or lower levels in malignancy.

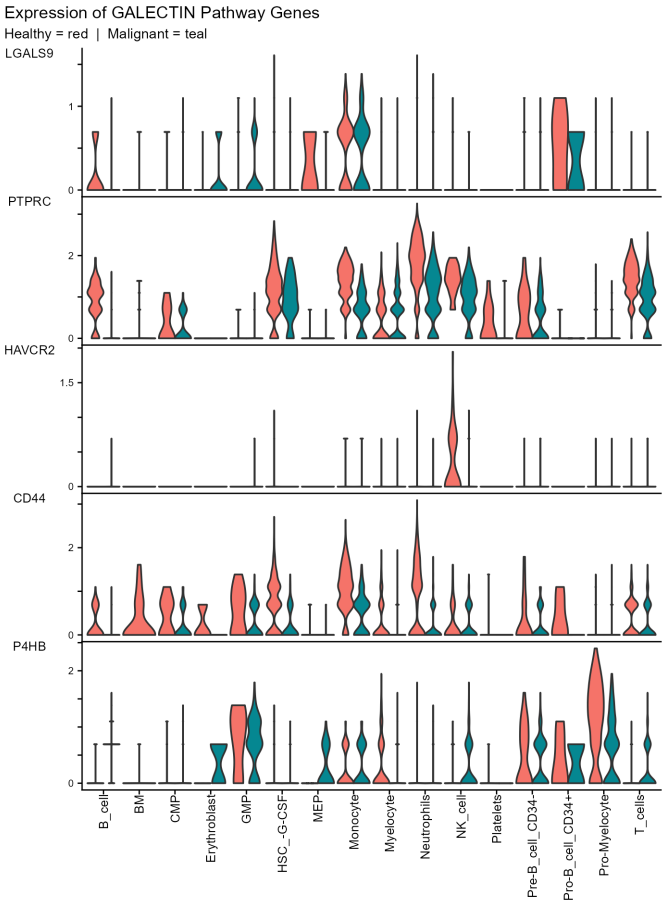


Figure39; *Violin plot Galectin pathway*

2.ANNEXIN Pathway Genes (Figure 40)

ANXA1 is highly expressed in Monocytes, Myelocytes, Neutrophils, and NK cells, with similar or lower expression in malignancy.

FPR1 and FPR2 are enriched in Myeloid lineages, with FPR2 lower in Malignant Myelocytes and Neutrophils, suggesting reduced chemotactic and inflammatory signaling.

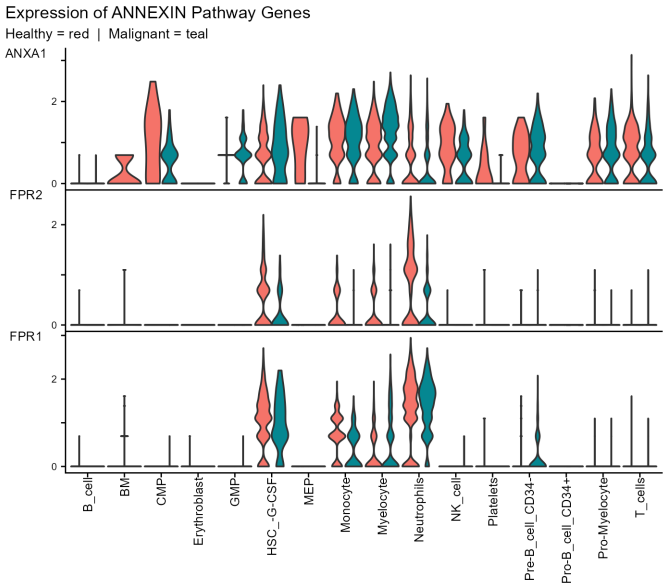


Figure40; *Violin plot Annexin pathway*

3.MIF Pathway Genes: (Figure41)

MIF and its receptor CD44 show lower expression in malignant samples across Monocytes, T cells, and Pro-B cells. CD74 is markedly lower in Malignant B cells and similar in T cells, so the MIF-CD74 axis is not reinforced in this sample. CXCR4 shows variable expression but is notably elevated in Malignant Monocytes and HSC-G-CSF cells, potentially enhancing cell migration or stromal interaction.

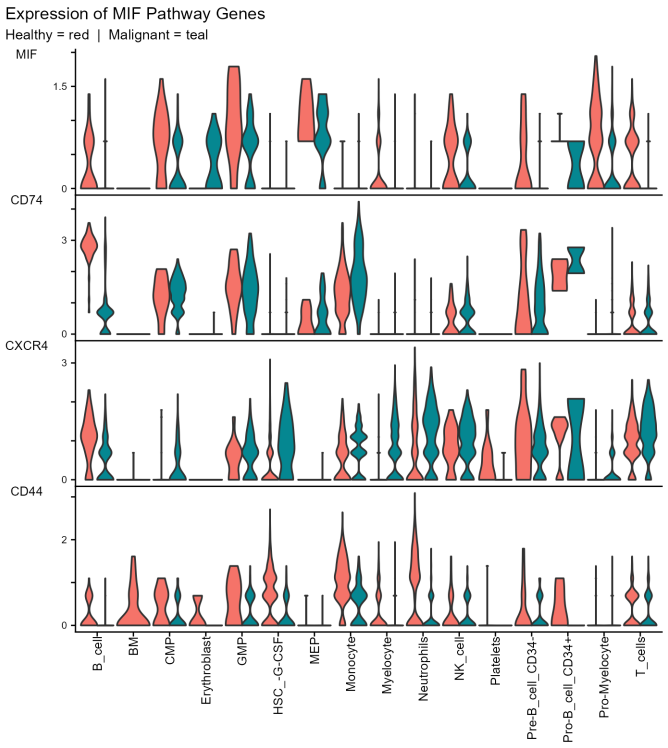


Figure41; *Violin plot MIF pathway*

IV. Biological Insights

1) Changes in Immunomodulatory Signaling (MIF, ANNEXIN, GALECTIN)

Analysis of the CellChat signaling data revealed a global reduction of cell-cell communication in malignant bone marrow (872 to 602 interactions; 41% lower interaction strength), within which the three pathways changed differently:

MIF Pathway: MIF signaling was reduced in the malignant condition, with about half the total communication probability of the healthy sample and a smaller share of the network (13% to 11%). Most cell-pair interactions were weaker, and MIF and CD74 expression was lower in most cell types.

ANNEXIN Pathway: ANNEXIN signaling was somewhat weaker in absolute terms but took a larger share of the remaining network (12% to 16%), centred on myeloid populations. Its immunosuppressive role may support the persistence of tumor-associated myeloid cells.

GALECTIN Pathway: GALECTIN signaling, sent mainly by monocytes, reached significance in more cell pairs in malignancy, including NK and T cells, while individual interactions were slightly weaker. GALECTIN is known to modulate T cell exhaustion and apoptosis, making this a candidate mechanism of immune evasion.

These patterns were consistent across bubble plots, circle plots, hierarchy plots, and violin plots of pathway gene expression.

2) Fibroblasts, Macrophages, and Plasma Cells

These cell types were not identified by the reference-based annotation used here, so their signaling roles could not be assessed directly; studying them would require a dedicated stromal or plasma-cell reference.

3) Proposed Mechanisms of Immune Evasion and Tumor Support

T Cell Exhaustion and Suppression: GALECTIN signaling from monocytes to T and NK cells can induce T cell dysfunction and apoptosis, weakening anti-tumor immunity.

Macrophage Polarization: The MIF-CD74 axis is reported to promote M2-like macrophage polarization; however, MIF signaling was reduced in this sample, so this mechanism is not supported by the present data.

Inhibitory Microenvironment: The larger share of ANNEXIN signaling may reduce pro-inflammatory signaling, limit immune activation, and enhance cell survival and stress responses in tumor-supportive cells.

Collectively, these findings point to a globally suppressed communication network in malignant bone marrow, within which ANNEXIN and GALECTIN signaling are relatively preserved and represent candidate immune-regulatory axes.